37

Wave Optics

CHAPTER OUTLINE

27.1 Young's Double-Slit Experiment

27.2 Analysis Model: Waves in Interference

27.3 Intensity Distribution of the Double-Slit Interference Pattern

27.4 Change of Phase Due to Reflection

27.5 Interference in Thin Films

27.6 The Michelson Interferometer

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ37.1 **(i)** Answer (a). If the mirrors do not move the character of the interference stays the same.

(ii) Answer (c). The light waves destructively interfere so they are initially out of phase by 180°. Moving the mirror by $\lambda/2$ changes the path difference by $2(\lambda/2) = \lambda$, so the waves go in phase then back out of phase to their original phase relation.

OQ37.2 **(i)** The ranking is b > a > c = d. The angles in the interference pattern are small, so we can make a good approximation of their values: $d\sin\theta = m\lambda \rightarrow \theta \approx m\lambda/d$. Thus for $m = 1$, $\theta \approx \lambda/d$, which we estimate in each case: (a) $0.450\ \mu\text{m}/400\ \mu\text{m} \approx 1.1 \times 10^{-3}$ rad (b) $0.7\ \mu\text{m}/400\ \mu\text{m} \approx 1.8 \times 10^{-3}$ rad (c) and (d) $0.7\ \mu\text{m}/800\ \mu\text{m} \approx 0.9 \times 10^{-3}$ rad.

(ii) The ranking is b = d > a > c. Now we consider the distance

$$y = L\tan\theta \approx L\sin\theta = L\left(m\lambda/d\right) \quad \rightarrow \quad y \approx mL\lambda/d$$

Thus for $m = 1$, $y \approx L\lambda/d$, which we estimate in each case:

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(a) (4 m) (0.45 μm/400 μm) $\approx$ 4.5 mm; (b) (4 m)(0.7 μm/400 μm) $\approx$ 7 mm; (c) (4 m)(0.7 μm/800 μm) $\approx$ 3.5 mm;
(d) (8 m)(0.7 μm/800 μm) $\approx$ 7 mm.

OQ37.3 Answer (c). Underwater, the wavelength of the light decreases according to $\lambda_{\text{water}} = \lambda_{\text{air}}/n_{\text{water}}$. Since the angles between positions of light and dark bands, being small, are approximately proportional to λ, the underwater fringe separations decrease.

OQ37.4 **(i)** Answer (c). The distance between nodes is half a wavelength.

(ii) Answer (d). The reflected light travels through the same path twice because it reflects, so moving the mirror one-quarter wavelength, 125 nm, results in a path change of one-half wavelength, 250 nm, which results in destructive interference.

(iii) Answer (e). The wavelength of the light in the film is 500 nm/2 = 250 nm. If the film is made 62.5 nm thicker (one-quarter wavelength in the film), the light reflecting inside the film has a path length 125 nm greater. This is half a wavelength, which reverses constructive into destructive interference.

OQ37.5 Answer (d). There are 180° phase changes occurring in the reflections at both the air-oil boundary and the oil-water boundary; thus the relative phase change from reflection is zero. The condition for constructive interference in the reflected light is

$$2t = m\frac{\lambda}{n} \quad \rightarrow \quad t = m\frac{\lambda}{2n}$$

where m is any integer. The minimum non-zero thickness of the oil which will strongly reflect 530-nm light is $m = 1$:

$$t = m\frac{\lambda}{2n} = (1)\frac{530 \text{ nm}}{2(1.25)} = 212 \text{ nm}$$

O37.6 Answer (a). For the second-order bright fringe,

$$d\sin\theta = 2\lambda$$

$$\sin\theta = 2\left(\frac{500\times10^{-9} \text{ m}}{2.00\times10^{-5} \text{ m}}\right)$$

$$\theta = 0.050\ 0 \text{ rad}$$

OQ37.7 **(i)** Answer (b). If the oil film is brightest where it is thinnest, then $n_{\text{air}} < n_{\text{oil}} < n_{\text{flint glass}}$. With this condition, light reflecting from both the top and the bottom surface of the oil film will undergo 180° phase changes. Then these two beams will be in phase with each other where the film is very thin. This is the condition for constructive interference as the thickness of the oil film

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decreases toward zero. If the oil film is dark where it is thinnest, then $n_{\text{air}} < n_{\text{oil}} > n_{\text{crown glass}}$. In this case, reflecting light undergoes a 180° phase change upon reflection from the top surface but no 180° phase change upon reflection from the bottom surface of the oil. The two reflected beams are 180° out of phase and interfere destructively as the oil film thickness goes to zero.

(ii) Yes. It should have a lower refractive index than both kinds of glass.

(iii) Yes. It should have a higher refractive index than both kinds of glass.

(iv) No. Its refractive index cannot be both greater than 1.66 and less than 1.52.

OQ37.8 Answer (b). With two fine slits separated by a distance d slightly less than λ, the equation $d\sin\theta = 0$ has the usual solution $\theta = 0$, but $d\sin\theta = \lambda$ has no solution: there is no first-order maximum.

However, $d\sin\theta = \frac{1}{2}\lambda$ has a solution: first-order minima flank the central maximum on each side.

OQ37.9 **(i)** Answer (a). The angular position of the mth-order bright fringe in a double-slit interference pattern is given by $d\sin\theta_m = m\lambda$. The distance y_m of the mth-order bright fringe from the center of the pattern is given by $y_m = L\tan\theta_m$, where L is the distance to the screen. The spacing between successive bright fringes is

$$\begin{aligned}\Delta y &= y_{m+1} - y_m = L\left(\tan\theta_{m+1} - \tan\theta_m\right)\\ &\approx L\left(\sin\theta_{m+1} - \sin\theta_m\right)\\ &= L\frac{\left[(m+1)-m\right]}{d}\lambda = \frac{L}{d}\lambda\end{aligned}$$

because the angles are small, and for small angles (in radians) $\sin\theta \simeq \tan\theta$. As L increases, the spacing Δy increases.

(ii) Answer (b). From our result above, we see that as d increases, the spacing Δy decreases.

OQ37.10 Answer (b). If the thickness of the oil film were smaller than half of the wavelengths of visible light, no colors would appear. If the thickness of the oil film were much larger, the colors would overlap to mix to white or gray.

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ37.1 A camera lens will have more than one element, to correct (at least) for chromatic aberration. It will have several surfaces, each of which would reflect some fraction of the incident light. To maximize light throughout, the surfaces need antireflective coatings. The coating thickness is chosen to produce destructive interference for reflected light of a particular wavelength.

CQ37.2 Due to gravity, the soap film tends to sag in its holder, being quite thin at the top and becoming thicker as one moves toward the bottom of the holding ring. Because light reflecting from the front surface of the film experiences a 180° phase change, and light reflecting from the back surface of the film does not (see Figure 37.10 in the textbook), the film must be a minimum of a half wavelength thick before it can produce constructive interference in the reflected light. Thus, the light must be striking the film at some distance from the top of the ring before the thickness is sufficient to produce constructive interference for any wavelength in the visible portion of the spectrum.

CQ37.3 The light from the flashlights consists of many different wavelengths (that's why it's white) with random time differences between the light waves. There is no *coherence* between the two sources. The light from the two flashlights does not maintain a constant phase relationship over time. These three equivalent statements mean no possibility of an interference pattern.

CQ37.4 Typically, a thin air film forms between the lens and the glass plate. Light reflecting from the upper surface of the air film (lower surface of the lens) can interfere with light reflecting from the lower surface of the air film (upper surface of the flat glass plate). The light reflecting from the *lower* surface of the air film undergoes a 180° phase change on reflection while the light reflecting from the *upper* surface of the air film does not. (a) Where there is negligible distance between the surfaces, at the center of the pattern you will see a dark spot because of the destructive interference associated with the 180° phase shift. (b) Colored rings surround the dark spot. If the lens is a perfect sphere and the plate is perfectly flat, the rings are perfect circles. On the fine scale of the wavelength of visible light, distorted rings reveal bumps and hollows that cause variation in the air film between the glass surfaces.

CQ37.5 The waves interfere destructively at some places and interfere constructively at others. The total energy is not lost, it is just rearranged. The energy that does not go into the dark fringes is shifted into the bright fringes.

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CQ37.6 Every color produces its own double-slit interference pattern, so if white light is used, the central maximum is white and the first-order maxima are full spectra running from violet to red. Each higher-order maximum is in principle a full spectrum, but it can partially overlap with the next order maximum, so the pattern for a specific color is hard to distinguish. Using monochromatic light eliminates this problem.

CQ37.7 (a) Two waves interfere constructively if their path difference is zero, or an integral multiple of the wavelength, according to $\delta = m\lambda,$ with $m = 0, 1, 2, 3,\ldots$.

(b) Two waves interfere destructively if their path difference is a half wavelength, or an odd multiple of $\frac{\lambda}{2},$ described by

$$\delta = \left(m + \frac{1}{2} \right) \lambda \text{, with } m = 0, 1, 2, 3,\ldots.$$

CQ37.8 Each liquid forms a film which causes interference of light reflected off the top and bottom surfaces of the film. Since the liquids would have an index greater than that of air, light reflected off the top surface of each film would undergo a 180° phase change. When the films become sufficiently thin, the type of interference that occurs, constructive or destructive, depends on whether the reflected wave does or does not undergo a 180° phase change. If the index of one liquid is less than that of water, light reflected off the bottom surface of the film (off the water surface) will be shifted by 180°, so the overall interference will be constructive, and the film will appear bright. If the index of the other liquid is greater than that of water, light reflected off the bottom surface of the film will not be shifted, so the overall interference will be destructive, and the film will appear dark.

CQ37.9 Yes. A single beam of laser light going into the slits divides up into several fuzzy-edged beams diverging from the point halfway between the slits.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 27.1 Young's Double-Slit Experiment

Section 27.2 Analysis Model: Waves in Interference

***P37.1** The angular locations of the bright fringes (or maxima) is given by Equation 37.2:

$$d\sin\theta = m\lambda$$

Solving for m and substituting 30.0° gives

$$m = \frac{d\sin\theta}{\lambda} = \frac{(3.20\times10^{-4}\ \text{m})\sin 30.0°}{500\times10^{-9}\ \text{m}} = 320$$

There are 320 maxima to the right, 320 to the left, and one for $m = 0$ straight ahead at $\theta = 0$. There are therefore $\boxed{641\ \text{maxima}}$.

P37.2 The location of the dark fringe of order m (measured from the position of the central maximum) is given by

$$(y_{\text{dark}})_m = \left(m+\frac{1}{2}\right)\left(\frac{L\lambda}{d}\right)$$

where $m = 0, \pm1, \pm2, \ldots$ Thus, the spacing between the first and second dark fringes will be

$$\Delta y = (y_{\text{dark}})_{m=1} - (y_{\text{dark}})_{m=0}$$
$$= \left(1+\frac{1}{2}\right)\left(\frac{L\lambda}{d}\right) - \left(0+\frac{1}{2}\right)\left(\frac{L\lambda}{d}\right) = \frac{L\lambda}{d}$$

or $$\Delta y = \frac{(5.30\times10^{-7}\ \text{m})(2.00\ \text{m})}{0.300\times10^{-3}\ \text{m}} = 3.53\times10^{-3}\ \text{m} = \boxed{3.53\ \text{mm}}$$

P37.3 The location of the bright fringe of order m (measured from the position of the central maximum) is

$$d\sin\theta = m\lambda \qquad m = 0, \pm1, \pm2, \ldots.$$

For first bright fringe to the side, $m = 1$. Thus, the wavelength of the laser light must be

$$\lambda = d\sin\theta = (0.200\times10^{-3}\text{m})\sin 0.181°$$
$$= 6.32\times10^{-7}\ \text{m} = \boxed{632\ \text{nm}}$$

P37.4 The location of the bright fringes for small angles is given by Equation 37.7:

$$y_{\text{bright}} = \frac{\lambda L}{d}m$$

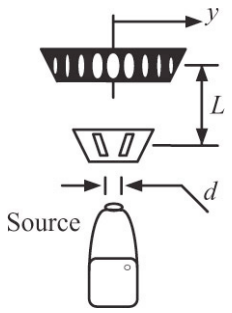
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For $m = 1$,

$$\lambda = \frac{y_{\text{bright}}}{L} = \frac{\left(3.40 \times 10^{-3}\ \text{m}\right)\left(0.500 \times 10^{-3}\ \text{m}\right)}{3.30\ \text{m}} = \boxed{515\ \text{nm}}$$

P37.5 In the equation $d \sin\theta = \left(m + \frac{1}{2}\right)\lambda$, the first minimum is described by $m = 0$ and the tenth by $m = 9$:

$$\sin\theta = \frac{\lambda}{d}\left(9 + \frac{1}{2}\right) = 9.5\frac{\lambda}{d}$$



ANS. FIG. P37.5

Also, $\tan\theta = \frac{y}{L}$. But, for small θ, $\sin\theta \approx \tan\theta$.

Thus, $d = \frac{9.5\lambda}{\sin\theta} = \frac{9.5\lambda L}{y}$:

$$d = \frac{9.5\left(5\ 890 \times 10^{-10}\ \text{m}\right)(2.00\ \text{m})}{7.26 \times 10^{-3}\ \text{m}} = 1.54 \times 10^{-3}\ \text{m} = \boxed{1.54\ \text{mm}}$$

P37.6 We use Equation 37.2, $d \sin\theta_{\text{bright}} = m\lambda$, to find the angle for the $m = 1$ fringe:

$$\sin\theta_{\text{bright}} = \frac{m\lambda}{d} = \frac{(1)\left(1.00 \times 10^{-2}\ \text{m}\right)}{8.00 \times 10^{-3}\ \text{m}} = 1.25$$

The sine of the angle is greater than 1, which is impossible. Therefore, there is no $m = 1$ fringe on the screen whose position can be measured. In fact, there is no interference pattern at all, just a bright area of microwaves directly behind the double slit.

P37.7 We do not use the small-angle approximation $\sin\theta \approx \tan\theta$ here because the angle is greater than 10°. For the first bright fringe, $m = 1$, and we have

$$d \sin\theta = m\lambda = \lambda$$

and

$$d = \frac{\lambda}{\sin\theta} = \frac{620 \times 10^{-9}\ \text{m}}{\sin 15.0^\circ} = 2.40 \times 10^{-6}\ \text{m} = \boxed{240\ \mu\text{m}}$$

P37.8 (a) For a bright fringe of order m, the path difference is $\delta = m\lambda$, where $m = 0, 1, 2, \ldots$ At the location of the third order bright fringe,

$$\delta = m\lambda = 3\left(589 \times 10^{-9}\ \text{m}\right) = 1.77 \times 10^{-6}\ \text{m} = \boxed{1.77\ \mu\text{m}}$$

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(b) For a dark fringe, the path difference is $\delta = \left(m + \frac{1}{2}\right)\lambda$, where $m = 0, 1, 2, \ldots$ At the third dark fringe, $m = 2$ and

$$\delta = \left(2 + \frac{1}{2}\right)\lambda = \frac{5}{2}(589 \text{ nm}) = 1.47 \times 10^3 \text{ nm} = \boxed{1.47\ \mu\text{m}}$$

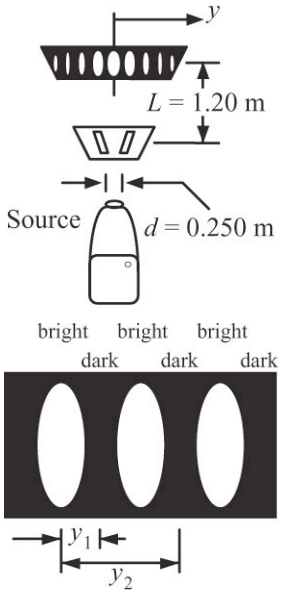
P37.9 (a) For the bright fringe,

$$y_{\text{bright}} = \frac{m\lambda L}{d}, \text{ where } m = 1$$

$$y = \frac{(546.1 \times 10^{-9}\ \text{m})(1.20\ \text{m})}{0.250 \times 10^{-3}\ \text{m}} = 2.62 \times 10^{-3}\ \text{m}$$

$$= \boxed{2.62\ \text{mm}}$$

(b) If you have trouble remembering whether the equation with $m\lambda$ or the equation with $\left(m + \frac{1}{2}\right)\lambda$ applies to a particular situation, you can remember that a zero-order bright band is in the center, and dark bands are halfway between bright bands. Thus, the made-up equation $d\sin\theta = (count)\lambda$ describes them all, with *count* = 0, 1, 2, … for bright bands, and with *count* = 0.5, 1.5, 2.5, … for dark bands.



ANS. FIG. P37.9

Then, for the dark bands,

$$y_{\text{dark}} = \frac{\lambda L}{d}\left(m + \frac{1}{2}\right);\ m = 0, 1, 2, 3, \ldots$$

$$\Delta y = y_2 - y_1 = \frac{\lambda L}{d}\left[\left(1 + \frac{1}{2}\right) - \left(0 + \frac{1}{2}\right)\right] = \frac{\lambda L}{d}$$

$$= \frac{(546.1 \times 10^{-9}\ \text{m})(1.20\ \text{m})}{0.250 \times 10^{-3}\ \text{m}}$$

$$\Delta y = \boxed{2.62\ \text{mm}}$$

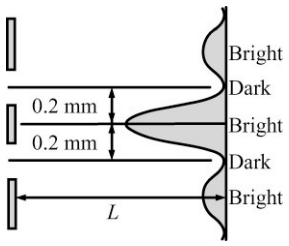
P37.10 Taking $m = 0$ and $y = 0.200$ mm in Equations 37.3 and 37.4 gives

$$L \approx \frac{2dy}{\lambda} = \frac{2(0.400 \times 10^{-3}\ \text{m})(0.200 \times 10^{-3}\ \text{m})}{442 \times 10^{-9}\ \text{m}} = 0.362\ \text{m}$$

$$L \approx \boxed{36.2\ \text{cm}}$$

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Geometric optics or a particle theory of light would incorrectly predict bright regions opposite the slits and darkness in between. But, as this example shows, interference can produce just the opposite.



ANS. FIG. P37.10

***P37.11** $\lambda = \dfrac{340\ \text{m/s}}{2\ 000\ \text{Hz}} = 0.170\ \text{m}$

The maxima are located at $d\sin\theta = m\lambda$:

$m = 0$ gives $\theta = 0°$

$m = 1$ gives $\theta = \sin^{-1}\left(\dfrac{\lambda}{d}\right) = \sin^{-1}\left(\dfrac{0.170\ \text{m}}{0.350\ \text{m}}\right) = 29.1°$

$m = 2$ gives $\theta = \sin^{-1}\left(\dfrac{2\lambda}{d}\right) = \sin^{-1}\left[\dfrac{2(0.170\ \text{m})}{0.350\ \text{m}}\right] = 76.3°$

$m = 3$ has no solution, since $\sin\theta > 1$.

The minima are located at $d\sin\theta = \left(m + \dfrac{1}{2}\right)\lambda$:

$m = 0$ gives $\theta = \sin^{-1}\left(\dfrac{\lambda}{2d}\right) = \sin^{-1}\left[\dfrac{0.170\ \text{m}}{2(0.350\ \text{m})}\right] = 14.1°$

$m = 1$ gives $\theta = \sin^{-1}\left(\dfrac{3\lambda}{2d}\right) = \sin^{-1}\left[\dfrac{3(0.170\ \text{m})}{2(0.350\ \text{m})}\right] = 46.8°$

$m = 2$ has no solution, since $\sin\theta > 1$.

We have maxima at 0°, 29.1°, and 76.3°; minima at 14.1° and 46.8°.

P37.12 The wavelength $\lambda = \dfrac{v}{f} = \dfrac{343\ \text{m/s}}{2\ 000\ \text{s}^{-1}} = 0.171\ 5\ \text{m}$ is on the same order of size as the slit separation $d = 0.300$ m, so we may treat this as a double-slit diffraction problem.

(a) $d\sin\theta = m\lambda$ so $(0.300\ \text{m})\sin\theta = 1(0.171\ 5\ \text{m})$ and $\theta =$ 34.9°.

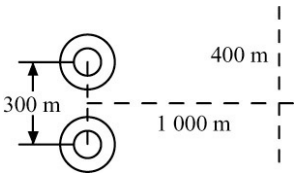
(b) $d\sin\theta = m\lambda$ so $d\sin 34.9° = 1(0.030\ 0\ \text{m})$ and $d =$ 5.25 cm.

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(c) $\left(1.00\times10^{-6}\text{ m}\right)\sin 34.9^\circ = (1)\lambda \quad \text{so} \quad \lambda = 572\text{ nm}.$

$$f = \frac{c}{\lambda} = \frac{3.00\times10^{8}\text{ m/s}}{5.72\times10^{-7}\text{ m}} = \boxed{5.24\times10^{14}\text{ Hz}}$$

P37.13 Note, with the conditions given, the small-angle approximation **does not work well**. That is, $\sin\theta$, $\tan\theta$, and θ are significantly different. We treat the interference as a Fraunhofer pattern.



ANS. FIG. P37.13

(a) At the $m = 2$ maximum,

$$\tan\theta = \frac{400\text{ m}}{1\,000\text{ m}} = 0.400 \rightarrow \theta = 21.8^\circ$$

So $$\lambda = \frac{d\sin\theta}{m} = \frac{(300\text{ m})\sin 21.8^\circ}{2} = \boxed{55.7\text{ m}}.$$

(b) The next minimum encountered is the $m = 2$ minimum, and at that point,

$$d\sin\theta = \left(m + \frac{1}{2}\right)\lambda$$

which becomes $d\sin\theta = \frac{5}{2}\lambda$,

or $$\sin\theta = \frac{5}{2}\frac{\lambda}{d} = \frac{5}{2}\left(\frac{55.7\text{ m}}{300\text{ m}}\right) = 0.464 \rightarrow \theta = 27.7^\circ,$$

so $$y = \left(1\,000\text{ m}\right)\tan 27.7^\circ = 524\text{ m}.$$

Therefore, the car must travel an additional

$$524\text{ m} - 400\text{ m} = \boxed{124\text{ m}}$$

If we considered Fresnel interference, we would more precisely find

(a) $\lambda = \frac{1}{2}\left(\sqrt{550^2 + 1\,000^2}\text{ m} - \sqrt{250^2 + 1\,000^2}\text{ m}\right) = 55.2\text{ m}$ and

(b) 123 m.

P37.14 Location of A = central maximum, location of B = first minimum.

So, $$\Delta y = \left[y_{\min} - y_{\max}\right] = \frac{\lambda L}{d}\left(0 + \frac{1}{2}\right) - 0 = \frac{1}{2}\frac{\lambda L}{d} = 20.0\text{ m}.$$

Thus, $$d = \frac{\lambda L}{2(20.0\text{ m})} = \frac{(3.00\text{ m})(150\text{ m})}{40.0\text{ m}} = \boxed{11.3\text{ m}}.$$

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P37.15 The angle θ of the 50th-order fringe is given by

$$d\sin\theta = m\lambda \;\rightarrow\; \theta = \sin^{-1}\left(\frac{50\lambda}{d}\right)$$

The distance x from the slit to the screen and the distance y of the mth-order fringe from the center of the central maximum are related by $\tan\theta = \frac{y}{x}$. As the student approaches the screen at speed v, the distances x and y decrease but their ratio stays the same. Therefore,

$$\tan\theta = \frac{y}{x} \quad\rightarrow\quad y = x\tan\theta$$

$$\frac{dy}{dt} = \frac{dx}{dt}\tan\theta = -v\tan\theta$$

where dy/dt is negative because the distance y shrinks. The *speed* of the fringe is

$$v_{50\text{th-order}} = \left|\frac{dy}{dt}\right| = v\tan\theta = v\tan\left[\sin^{-1}\left(\frac{m\lambda}{d}\right)\right]$$

Thus, the speed of the 50th-order fringe is

$$v_{50\text{th-order}} = (3.00\text{ m/s})\tan\left\{\sin^{-1}\left[\frac{50\left(632.8\times10^{-9}\text{ m}\right)}{0.300\times10^{-3}\text{ m}}\right]\right\}$$

$$= \boxed{0.318\text{ m/s}}$$

P37.16 The angle θ of the mth-order fringe is given by

$$d\sin\theta = m\lambda \quad\rightarrow\quad \theta = \sin^{-1}\left(\frac{m\lambda}{d}\right)$$

The distance x from the slit to the screen and the distance y of the mth-order fringe from the center of the central maximum are related by $\tan\theta = \frac{y}{x}$. As the student approaches the screen at speed v, the distances x and y decrease but their ratio stays the same. Therefore,

$$\tan\theta = \frac{y}{x} \quad\rightarrow\quad y = x\tan\theta$$

$$\frac{dy}{dt} = \frac{dx}{dt}\tan\theta = -v\tan\theta$$

where dy/dt is negative because the distance y shrinks. Thus, the speed of the mth-order fringe is

$$v_{m\text{th-order}} = \left|\frac{dy}{dt}\right| = v\tan\theta = \boxed{v\tan\left[\sin^{-1}\left(\frac{m\lambda}{d}\right)\right]}$$

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P37.17 As shown in the figure to the right, the height of the radio telescope dish is $h = d_2 \sin\theta$, and the path difference in the waves reaching the telescope is

$$\delta = d_2 - d_1 = d_2(1 - \sin\alpha)$$

where

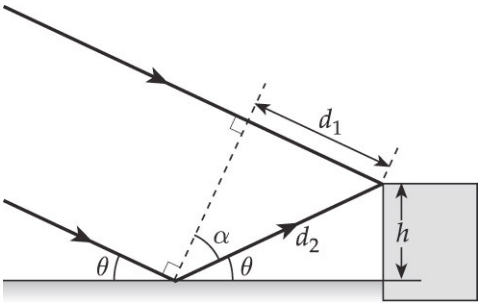
$$\theta + \alpha + \theta = 90° \quad \rightarrow \quad \alpha = 90° - 2\theta$$

If the first minimum $(\delta = \lambda/2)$ occurs when $\theta = 25.0°$, then

$$\alpha = 90° - 2(25.0°) = 40.0°\text{, and}$$

$$d_2 = \frac{\delta}{1 - \sin\alpha} = \frac{(250\text{ m})/2}{1 - \sin 40.0°} = 350\text{ m}$$

Thus, the height $h = d_2 \sin\theta = 350\text{ m} \sin 25.0° =$ $\boxed{148\text{ m}}$



ANS. FIG. P37.17

P37.18 For a double-slit system, the path difference of the two wave fronts arriving at a screen is $\delta = d \sin\theta$ and the phase difference is

$$\phi = \frac{2\pi}{\lambda}\delta = \frac{2\pi}{\lambda} d \sin\theta \approx \frac{2\pi}{\lambda} d\left(\frac{y}{L}\right)$$

(a) For $\theta = 0.500°$,

$$\phi = \frac{2\pi}{\lambda} d \sin\theta$$

$$\phi = \frac{2\pi}{(500 \times 10^{-9}\text{ m})}(0.120 \times 10^{-3}\text{ m})\sin(0.500°) = \boxed{13.2\text{ rad}}$$

(b) $$\phi \approx \frac{2\pi}{\lambda} d\left(\frac{y}{L}\right) = \frac{2\pi}{(500 \times 10^{-9}\text{ m})}(0.120 \times 10^{-3}\text{ m})\left(\frac{5.00 \times 10^{-3}\text{ m}}{1.20\text{ m}}\right)$$

$$= \boxed{6.28\text{ rad}}$$

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(c) If $\phi = 0.333 \text{ rad} = \dfrac{2\pi d \sin\theta}{\lambda}$, then

$$\theta = \sin^{-1}\left(\frac{\lambda\phi}{2\pi d}\right) = \sin^{-1}\left[\frac{(500\times10^{-9}\text{ m})(0.333\text{ rad})}{2\pi(0.120\times10^{-3}\text{ m})}\right]$$

$$\theta = \boxed{1.27\times10^{-2}\,^\circ}$$

(d) If $d\sin\theta = \dfrac{\lambda}{4}$, then

$$\theta = \sin^{-1}\left(\frac{\lambda}{4d}\right) = \sin^{-1}\left[\frac{500\times10^{-9}\text{ m}}{4(0.120\times10^{-3}\text{ m})}\right]$$

$$\theta = \boxed{5.97\times10^{-2}\,^\circ}$$

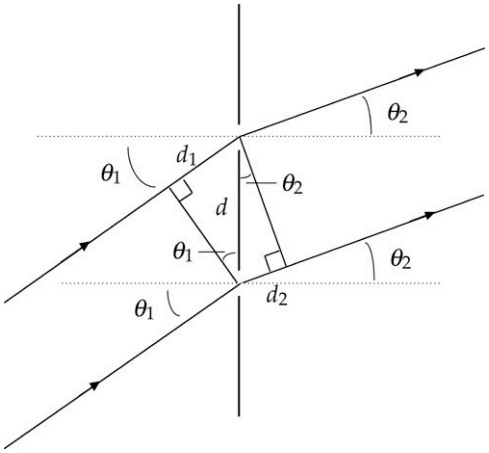
P37.19 From the diagram, the path difference between rays 1 and 2 is

$$\delta = d_1 - d_2 = d\sin\theta_1 - d\sin\theta_2$$

For constructive interference, this path difference must be equal to an integral number of wavelengths:

$$d\sin\theta_1 - d\sin\theta_2 = m\lambda$$

$$\sin\theta_1 - \sin\theta_2 = \frac{m\lambda}{d} \quad\rightarrow\quad \theta_2 = \sin^{-1}\left(\sin\theta_1 - \frac{m\lambda}{d}\right)$$



ANS. FIG. P37.19

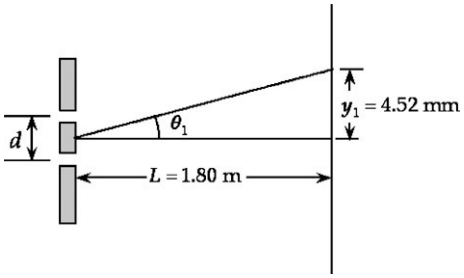
P37.20 (a) $y = 50y_{\text{bright}} = 50\left(4.52\times10^{-3}\text{ m}\right) = 0.226\text{ m} = \boxed{22.6\text{ cm}}$

(b) $\tan\theta_1 = \dfrac{\left(y_{\text{bright}}\right)_{m=1}}{L} = \dfrac{4.52\times10^{-3}\text{ m}}{1.80\text{ m}} = \boxed{2.51\times10^{-3}}$

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(c) From (b), $\theta_1 = \tan^{-1}\left(\dfrac{4.52\times10^{-3}\ \text{m}}{1.80\ \text{m}}\right) = 0.144°$

$\rightarrow \quad \sin\theta_1 = 2.51\times10^{-3}$



ANS. FIG. P37.20

The sine and the tangent are very nearly the same, but only because the angle is small. From $d\sin\theta_{\text{bright}} = m\lambda$, for $m = 1$:

$$\lambda = \frac{d\sin\theta_1}{1} = \frac{\left(2.40\times10^{-4}\ \text{m}\right)\sin\left(0.144°\right)}{1} = \boxed{6.03\times10^{-7}\ \text{m}}$$

(d) From $\delta = d\sin\theta = m\lambda$ for the order m bright fringe,

$$\theta_{50} = \sin^{-1}\left(\frac{50\lambda}{d}\right) = \sin^{-1}\left(50\sin\theta_1\right) = \sin^{-1}\left[50\sin\left(0.144°\right)\right]$$

$$= \boxed{7.21°}$$

(e) $y_5 = L\tan\theta_5 = \left(1.80\ \text{m}\right)\tan\left(7.21°\right) = 2.26\times10^{-2}\ \text{m} = \boxed{2.28\ \text{cm}}$

(f) The two answers are close but do not agree exactly. The fringes are not laid out linearly on the screen as assumed in part (a), and this nonlinearity is evident for relatively large angles such as 7.21°.

P37.21 (a) The path difference $\delta = d\sin\theta$, and when $L >> y$:

$$\delta = \frac{yd}{L} = \frac{\left(1.80\times10^{-2}\ \text{m}\right)\left(1.50\times10^{-4}\ \text{m}\right)}{1.40\ \text{m}}$$

$$= 1.93\times10^{-6}\ \text{m} = \boxed{1.93\ \mu\text{m}}$$

(b) $\dfrac{\delta}{\lambda} = \dfrac{1.93\times10^{-6}\ \text{m}}{6.43\times10^{-7}\ \text{m}} = 3.00$, or $\boxed{\delta = 3.00\lambda}$

(c) Point P will be a $\boxed{\text{maximum}}$ because the path difference is an integer multiple of the wavelength.

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P37.22 Observe that the pilot must not only home in on the airport, but must be headed in the right direction when she arrives at the end of the runway.

(a) $\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{30.0 \times 10^6 \text{ s}^{-1}} = \boxed{10.0 \text{ m}}$

(b) The first side maximum is at an angle given by $d \sin\theta = (1)\lambda$.

$$(40.0 \text{ m}) \sin\theta = 10.0 \text{ m} \qquad \theta = 14.5°$$

The 2.00 km is the length of the hypotenuse of a triangle with angle θ:

$$y = L \sin\theta = (2\,000 \text{ m}) \sin 14.5° = \boxed{500 \text{ m}}$$

(c) The intent is to inform the pilot which signal corresponds to the central maximum. The signal of 10-m wavelength in parts (a) and (b) would show maxima at 0°, 14.5°, 30.0°, 48.6°, and 90°. A signal of wavelength, say, 11.23 m, would show maxima at 0°, 16.3°, 34.2°, and 57.3°. The only value in common is 0°. A strong signal for both frequencies would indicate that the airplane was traveling along the central maximum, thus, straight on the runway. If λ_1 and λ_2 were related by a ratio of small integers in $\frac{\lambda_1}{\lambda_2} = \frac{n_1}{n_2}$, equivalent to $\frac{f_2}{f_1} = \frac{n_1}{n_2}$, then the equations $d \sin\theta = n_2 \lambda_1$ and $d \sin\theta = n_1 \lambda_2$ would both be satisfied for the same nonzero angle. The pilot could approach on an inappropriate bearing, and run off the runway immediately after touchdown.

Section 37.3 Intensity Distribution of the Double-Slit Interference Pattern

P37.23 We use Equation 37.14,

$$I = I_{\max} \cos^2\left(\frac{\pi y d}{\lambda L}\right)$$

Solving and substituting then gives

$$\frac{I}{I_{\max}} = \cos^2\left[\frac{\pi\left(6.00 \times 10^{-3} \text{ m}\right)\left(1.80 \times 10^{-4} \text{ m}\right)}{\left(656.3 \times 10^{-9} \text{ m}\right)(0.800 \text{ m})}\right] = \boxed{0.968}$$

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P37.24 We use trigonometric identities to write

$$E_1 + E_2 = 6.00 \sin(100\pi t) + 8.00 \sin(100\pi t + \pi/2)$$

$$= 6.00 \sin(100\pi t) + \left[8.00 \sin(100\pi t)\cos(\pi/2) + 8.00 \cos(100\pi t)\sin(\pi/2)\right]$$

$$E_1 + E_2 = 6.00 \sin(100\pi t) + 8.00 \cos(100\pi t)$$

and

$$E_R \sin(100\pi t + \phi) = E_R \sin(100\pi t)\cos\phi + E_R \cos(100\pi t)\sin\phi$$

The equation $E_1 + E_2 = E_R \sin(100\pi t + \phi)$ is satisfied if we require

$$600 = E_R \cos\phi \quad \text{and} \quad 8.00 = E_R \sin\phi$$

or $$(6.00)^2 + (8.00)^2 = E_R{}^2(\cos^2\phi + \sin^2\phi) \rightarrow \boxed{E_R = 10.0}$$

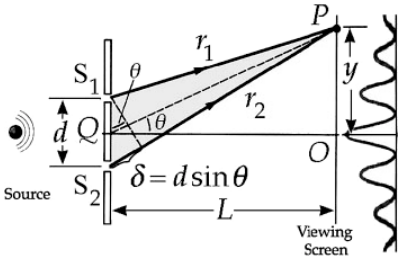
and $$\tan\phi = \sin\phi/\cos\phi = 8.00/6.00 = 1.33 \rightarrow \boxed{\phi = 53.1^\circ}$$

P37.25 We will use Equation 37.14 for intensity in a double-slit interference pattern, which is

$$I = I_{\max} \cos^2\left[\frac{\pi d \sin\theta}{\lambda}\right]$$

For small θ, from ANS. FIG. P37.25,

$$\sin\theta \approx \frac{y}{L}$$



ANS. FIG. P37.25

Substituting and solving gives

$$y = \frac{\lambda L}{\pi d}\cos^{-1}\sqrt{\frac{I}{I_{\max}}}$$

Next, with $I = 0.750 I_{\max}$, we can substitute a value for each variable:

$$y = \frac{(6.00 \times 10^{-7}\ \text{m})(1.20\ \text{m})}{\pi(2.50 \times 10^{-3}\ \text{m})}\cos^{-1}\sqrt{0.750} = \boxed{48.0\ \mu\text{m}}$$

P37.26 (a) The resultant amplitude is

$$E_r = E_0 \sin\omega t + E_0 \sin(\omega t + \phi) + E_0 \sin(\omega t + 2\phi)$$

where $\phi = \frac{2\pi}{\lambda} d \sin\theta$.

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Expanding,

$$E_r = E_0\left(\sin\omega t + \sin\omega t\cos\phi + \cos\omega t\sin\phi + \sin\omega t\cos 2\phi + \cos\omega t\sin 2\phi\right)$$

$$E_r = E_0\left(\sin\omega t\right)\left(1+\cos\phi+2\cos^2\phi-1\right) + E_0\left(\cos\omega t\right)\left(\sin\phi+2\sin\phi\cos\phi\right)$$

$$E_r = E_0\left(1+2\cos\phi\right)\left(\sin\omega t\cos\phi+\cos\omega t\sin\phi\right)$$

$$= E_0\left(1+2\cos\phi\right)\sin\left(\omega t+\phi\right)$$

Then the intensity is

$$I \propto E_r^2 = E_0^2\left(1+2\cos\phi\right)^2\left(\frac{1}{2}\right)$$

where we have substituted the time average of $\sin^2\left(\omega t+\phi\right)$, which is $\frac{1}{2}$. The maximum intensity occurs at $\phi = 0$:

$$I_{\max} \propto E_0^2\left(1+2\cos 0\right)^2\left(\frac{1}{2}\right) = \frac{9}{2}E_0^2$$

Therefore, the ratio of intensity to maximum intensity is

$$\frac{I}{I_{\max}} = \frac{E_0^2\left(1+2\cos\phi\right)^2\left(\frac{1}{2}\right)}{\frac{9}{2}E_0^2} = \frac{\left(1+2\cos\phi\right)^2}{9}$$

$$I = \frac{I_{\max}}{9}\left(1+2\cos\phi\right)^2$$

$$\boxed{I = \frac{I_{\max}}{9}\left[1+2\cos\left(\frac{2\pi d\sin\theta}{\lambda}\right)\right]^2}$$

(b) Look at the $N = 3$ graph in the textbook Figure 37.7. The intensity is zero at two places between the relative maxima, attained where $\cos\phi = -\frac{1}{2}$. The relative secondary maximum in the middle occurs at $\cos\phi = -1.00$, where $I = \frac{I_{\max}}{9}[1-2]^2 = \frac{I_{\max}}{9}$.

(c) The larger local maximum happens where $\cos\phi = +1.00$, giving $I = \frac{I_{\max}}{9}[1+2]^2 = I_{\max}$. The ratio of intensities at primary versus secondary maxima is $\boxed{9:1}$.

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P37.27 (a) From Equation 37.14,

$$I = I_{\max}\cos^2\left(\frac{\pi d\sin\theta}{\lambda}\right)$$

with $\phi = \frac{2\pi}{\lambda}d\sin\theta$. This gives

$$\frac{I}{I_{\max}} = \cos^2\left(\frac{\phi}{2}\right)$$

Therefore,

$$\phi = 2\cos^{-1}\sqrt{\frac{I}{I_{\max}}} = 2\cos^{-1}\sqrt{0.640} = \boxed{1.29 \text{ rad}}$$

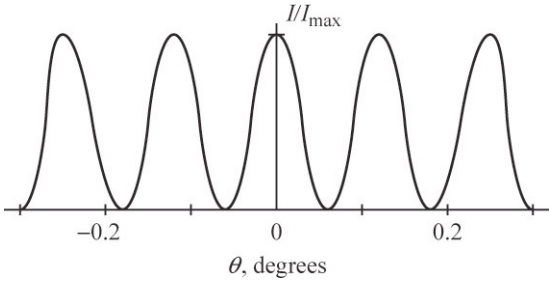
P37.28 In $I_{\text{avg}} = I_{\max}\cos^2\left(\frac{\pi d\sin\theta}{\lambda}\right)$ for angles between −0.3° and +0.3° we may take $\sin\theta = \theta$ (in radians) to find

$$I = I_{\max}\cos^2\left[\frac{\pi(250\ \mu\text{m})\ \theta}{0.546\ \mu\text{m}}\right]$$

This equation is correct assuming θ is in radians; but we can then equally well substitute in values for θ in degrees and interpret the argument of the cosine function as a number of degrees. We get the same answers for θ negative and for θ positive. We evaluate

θ degrees	−0.30	−0.25	−0.20	−0.15	−0.10	−0.05	0.0
$I/I_{\max}$	0.101	1.00	0.092	0.659	0.652	0.096	1.00
θ degrees	0.05	0.10	0.15	0.20	0.25	0.30	
$I/I_{\max}$	0.096	0.652	0.659	0.092	1.00	0.101	

TABLE P37.28



ANS. FIG. P37.28

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The cosine-squared function has maximum values of 1 at $\theta = 0$, at $\theta = 0.125°$, and at $\theta = 0.250°$. It has minimum values of zero halfway between the maximum values. The graph then has the appearance shown.

P37.29 (a) From Equation 37.9,

$$\phi = \frac{2\pi d}{\lambda}\sin\theta = \frac{2\pi d}{\lambda}\cdot\frac{y}{\sqrt{y^2 + D^2}}$$

$$\phi \approx \frac{2\pi yd}{\lambda D} = \frac{2\pi\left(0.850\times10^{-3}\text{ m}\right)\left(2.50\times10^{-3}\text{ m}\right)}{\left(600\times10^{-9}\text{ m}\right)(2.80\text{ m})} = \boxed{7.95\text{ rad}}$$

(b) $$\frac{I}{I_{\max}} = \frac{\cos^2\left[\left(\pi d/\lambda\right)\sin\theta\right]}{\cos^2\left[\left(\pi d/\lambda\right)\sin\theta_{\max}\right]} = \frac{\cos^2\left(\phi/2\right)}{\cos^2 m\pi}$$

$$\frac{I}{I_{\max}} = \cos^2\frac{\phi}{2} = \cos^2\left(\frac{7.95\text{ rad}}{2}\right) = \boxed{0.453}$$

Section 37.4 Change of Phase Due to Reflection

Section 37.5 Interference in Thin Films

P37.30 (a) With phase reversal in the reflection at the outer surface of the soap film and no reversal on reflection from the inner surface, the condition for constructive interference in the light reflected from the soap bubble is

$$2t = \left(m + \frac{1}{2}\right)\lambda_n = \left(m + \frac{1}{2}\right)\frac{\lambda}{n} \quad\rightarrow\quad 2nt = \left(m + \frac{1}{2}\right)\lambda$$

$$\lambda = \frac{2nt}{\left(m + \frac{1}{2}\right)}$$

where $m = 0, 1, 2, \ldots$. For the lowest order reflection ($m = 0$), and the wavelength is

$$\lambda = \frac{2nt}{\left(0 + 1/2\right)} = \frac{2(1.33)(120\text{ nm})}{1/2} = \boxed{638\text{ nm}}$$

(b) A thicker film would require a higher order of reflection, so use a larger value of m.

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(c) From (a) above, for a given wavelength, the thickness would be

$$t = \left(m + \frac{1}{2}\right)\frac{\lambda}{2n} = \left(m + \frac{1}{2}\right)\frac{638\text{ nm}}{2(1.33)}$$

The next greater thickness of soap film that can strongly reflect 638 nm light corresponds to $m = 1$, giving

$$t = \left(m + \frac{1}{2}\right)\frac{\lambda}{2n} = \left(1 + \frac{1}{2}\right)\frac{638\text{ nm}}{2(1.33)} = \boxed{360\text{ nm}}$$

and the third such thickness (corresponding to $m = 2$) is

$$t = \left(m + \frac{1}{2}\right)\frac{\lambda}{2n} = \left(2 + \frac{1}{2}\right)\frac{638\text{ nm}}{2(1.33)} = \boxed{600\text{ nm}}$$

P37.31 The layers are air, oil, and water. Because $1 < 1.25 < 1.33$, light reflected both from the top and from the bottom surface of the oil suffers phase reversal. For constructive interference we require

$$2t = \frac{m\lambda_{\text{cons}}}{n}$$

and for destructive interference,

$$2t = \frac{\left[m + (1/2)\right]\lambda_{\text{des}}}{n}$$

Then, $$\frac{\lambda_{\text{cons}}}{\lambda_{\text{dest}}} = 1 + \frac{1}{2m} = \frac{640\text{ nm}}{512\text{ nm}} = 1.25 \text{ and } m = 2$$

Therefore, $t = \dfrac{2(640\text{ nm})}{2(1.25)} = \boxed{512\text{ nm}}$.

P37.32 There are a total of two phase reversals caused by reflection, one at the top and one at the bottom surface of the coating.

$$2nt = \left(m + \frac{1}{2}\right)\lambda \quad \text{so} \quad t = \left(m + \frac{1}{2}\right)\frac{\lambda}{2n}$$

The minimum thickness of the film is therefore

$$t = \left(\frac{1}{2}\right)\frac{(500\text{ nm})}{2(1.30)} = \boxed{96.2\text{ nm}}$$

P37.33 Treating the anti-reflectance coating like a camera-lens coating (two phase reversals caused by reflection, one at the top and one at the bottom surface of the coating),

$$2t = \left(m + \frac{1}{2}\right)\frac{\lambda}{n} \quad \rightarrow \quad 2nt = \left(m + \frac{1}{2}\right)\lambda$$

(destructive interference)

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Let $m = 0$. Then,

$$t = \frac{\lambda}{4n} = \frac{3.00 \text{ cm}}{4(1.50)} = \boxed{0.500 \text{ cm}}$$

This anti-reflectance coating could be easily countered by changing the wavelength of the radar to 1.50 cm. Then the coating would exhibit maximum reflection!

P37.34 (a) The film thickness is $t = 1.00 \times 10^{-5}$ cm $= 1.00 \times 10^{-7}$ m $= 100$ nm. Since the light undergoes a 180° phase change at each surface of the film, the condition for *constructive* interference is

$$2t = m\frac{\lambda}{n}, \quad \text{or} \quad \lambda = \frac{2nt}{m} = \frac{2(1.38)(100 \text{ nm})}{m} = \frac{276 \text{ nm}}{m}$$

Therefore, the wavelengths intensified in the reflected light are, for $m = 1$, 2, and 3:

$$\lambda = \boxed{276 \text{ nm},\ 138 \text{ nm},\ 92.0 \text{ nm}}$$

(b) $\boxed{\text{No visible wavelengths are intensified.}}$ Because $m \geq 1$, all reflection maxima are in the ultraviolet and beyond.

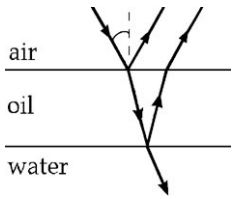
P37.35 If the path length difference $\Delta = \lambda$, the transmitted light will be bright. Since $\Delta = 2d = \lambda$,

$$d_{\min} = \frac{\lambda}{2} = \frac{580 \text{ nm}}{2} = \boxed{290 \text{ nm}}$$

P37.36 (a) The light reflected from the top of the oil film undergoes phase reversal. Since 1.45 > 1.33, the light reflected from the bottom undergoes no reversal. For constructive interference of reflected light, we then have

$$2t = \left(m + \frac{1}{2}\right)\frac{\lambda}{n} \rightarrow 2nt = \left(m + \frac{1}{2}\right)\lambda$$

or $$\lambda_m = \frac{2nt}{m + 1/2} = \frac{2(1.45)(280 \text{ nm})}{m + 1/2} = \frac{812 \text{ nm}}{m + 1/2}.$$



ANS. FIG. P37.36

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Substituting for m gives:

$m = 0,\ \lambda_0 = 1\ 620\ \text{nm}$ (infrared)

$m = 1,\ \lambda_1 = 541\ \text{nm}$ (green)

$m = 2,\ \lambda_2 = 325\ \text{nm}$ (ultraviolet)

Both infrared and ultraviolet light are invisible to the human eye, so the dominant color in reflected light is **green**.

(b) The dominant wavelengths in the transmitted light are those that produce destructive interference in the reflected light. The condition for destructive interference upon reflection is

$$2t = m\frac{\lambda}{n}$$

or $$\lambda_m = \frac{2nt}{m} = \frac{812\ \text{nm}}{m}.$$

Substituting for m gives:

$m = 1,\ \lambda_1 = 812\ \text{nm}$ (near infrared)

$m = 2,\ \lambda_2 = 406\ \text{nm}$ (violet)

$m = 3,\ \lambda_3 = 271\ \text{nm}$ (ultraviolet)

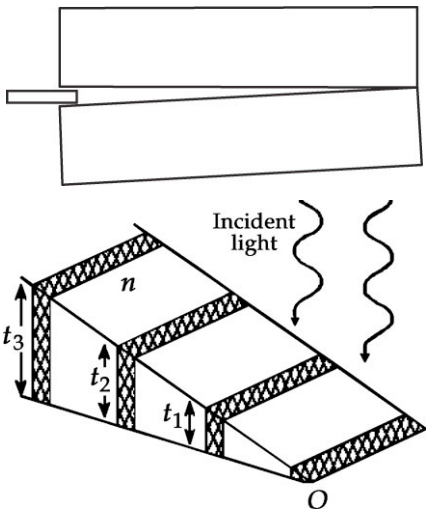
Of these, the only wavelength visible to the human eye (and hence the dominant wavelength observed in the transmitted light) is 406 nm. Thus, the dominant color in the transmitted light is **violet**.

P37.37 For destructive interference in the air,

$$2t = m\lambda$$

The first dark fringe occurs at the end where the plates meet, where destructive interference occurs because of the phase reversal caused by light reflecting from the top of the lower glass slide. For 30 dark fringes, including the one where the plates meet, $m = 29$ and

$$t = \frac{n\lambda}{2} = \frac{29(600\ \text{nm})}{2}$$
$$= 8.70 \times 10^{-6}\ \text{m} = 8.70\ \mu\text{m}$$



ANS. FIG. P37.37

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The *diameter* of the wire is the same as the thickness:

$$d = t = \boxed{8.70\ \mu\text{m}}$$

P37.38 Light waves are partially reflected and transmitted by the partially aluminized glass surfaces on the front and back surfaces of the filter. For maximum transmission, we want destructive interference between the waves reflected from the front and back surfaces of the film: the result of this interference is that most light of the H_α line is transmitted through the filter.

(a) If the surrounding glass has refractive index greater than 1.378, light reflected from the front surface of the filter (glass-filter interface) suffers no phase reversal and light reflected from the back surface of the filter (filter-glass interface) does undergo phase reversal. This effect by itself would produce destructive interference, so we want the distance down and back to be one whole wavelength in the film: $2t = \dfrac{\lambda}{n}$.

$$t = \frac{\lambda}{2n} = \frac{656.3\ \text{nm}}{2(1.378)} = \boxed{238\ \text{nm}}$$

(b) The filter will undergo thermal expansion. As t increases in $2nt = \lambda$, so does $\boxed{\lambda \text{ increase}}$.

(c) Destructive interference for reflected light happens when $2t = \dfrac{2\lambda}{n}$:

$$\lambda = nt = 1.378(238\ \text{nm}) = \boxed{328\ \text{nm}}\ \text{(near ultraviolet)}$$

P37.39 Reflection off the lower glass plate causes a phase reversal. The condition for bright fringes is

$$2t = \left(m + \frac{1}{2}\right)\frac{\lambda}{n} \qquad m = 0,\ 1,\ 2,\ 3,\ \ldots.$$

From ANS. FIG. P37.39, observe that

$$t = R(1 - \cos\theta) \approx R\left(1 - 1 + \frac{\theta^2}{2}\right) = \frac{R}{2}\left(\frac{r}{R}\right)^2 = \frac{r^2}{2R}$$

The condition for a bright fringe becomes

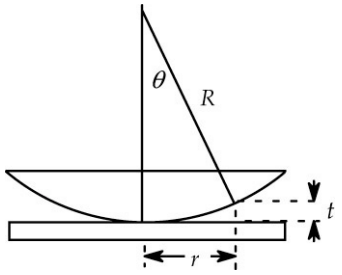
$$\frac{r^2}{R} = \left(m + \frac{1}{2}\right)\frac{\lambda}{n}$$

Thus, for fixed m and λ, nr^2 = constant.

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Therefore,

$$n_{\text{liquid}} r_{fi}^2 = n_{\text{air}} r^2 \quad \text{and} \quad n_{\text{liquid}} = (1.00)\frac{(1.50\text{ cm})^2}{(1.31\text{ cm})^2} = \boxed{1.31}$$



ANS. FIG. P37.39

P37.40 (a) The missing wavelength in reflected light is caused by destructive interference. The index of the coating (1.38) is greater than that of air (1.00), and the index of the glass (1.52) is greater than that of the coating; therefore, light waves reflected off the front and back surfaces of the coating undergo phase reversals. For destructive interference,

$$2t = \left(m + \frac{1}{2}\right)\frac{\lambda}{n} \qquad m = 0, 1, 2, 3, \ldots \quad \text{and} \quad n = 1.38$$

For the minimum thickness, $m = 0$:

$$2t = \left(m + \frac{1}{2}\right)\frac{\lambda}{n} \quad \rightarrow \quad t = \frac{\lambda}{4n} = \frac{540\text{ nm}}{4(1.38)} = \boxed{97.8\text{ nm}}$$

(b) Yes. Destructive interference occurs when $2nt = (m + \frac{1}{2})\lambda$ (Eq. 37.17), where m is an integer. (There is a phase change at both faces of the film in Figure P37.40.) Hence, for $m =$ 1, 2, ... we obtain thicknesses of 293 nm, 489 nm, ...

P37.41 For total darkness, we want destructive interference for reflected light for both 400 nm and 600 nm. With phase reversal at just one reflecting surface (the bottom glass plate), the condition for destructive interference is

$$2n_{\text{air}} t = m\lambda \qquad m = 0, 1, 2, \ldots .$$

The least common multiple of these two wavelengths is 1 200 nm, so we get no reflected light at $2(1.00)t = 3(400\text{ nm}) = 2(600\text{ nm}) = 1\,200$ nm, so $t = 600$ nm at this second dark fringe.

By similar triangles, $\dfrac{600\text{ nm}}{x} = \dfrac{0.050\,0\text{ mm}}{10.0\text{ cm}}$

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or the distance from the contact point is

$$x = \left(600 \times 10^{-9}\ \text{m}\right)\left(\frac{0.100\ \text{m}}{5.00 \times 10^{-5}\ \text{m}}\right) = \boxed{1.20\ \text{mm}}$$

Section 37.6 The Michelson Interferometer

P37.42 When the mirror on one arm is displaced by $\Delta\ell$, the path difference changes by $2\Delta\ell$. A shift resulting in the reversal between dark and bright fringes requires a path length change of one-half wavelength. Therefore, $2\Delta\ell = \frac{m\lambda}{2}$, where in this case, $m = 250$.

$$\Delta\ell = m\frac{\lambda}{4} = \frac{(250)\left(6.328 \times 10^{-7}\ \text{m}\right)}{4} = \boxed{39.6\ \mu\text{m}}$$

***P37.43** Counting light going both directions, the number of wavelengths originally in the cylinder is $m_1 = \frac{2L}{\lambda}$. It changes to $m_2 = \frac{2L}{\lambda/n_{\text{gas}}} = \frac{2n_{\text{gas}}L}{\lambda}$ as the cylinder is filled with gas. If N is the number of bright fringes passing, $N = m_2 - m_1 = \frac{2L}{\lambda}\left(n_{\text{gas}} - 1\right)$, or the index of refraction of the gas is

$$n_{\text{gas}} = 1 + \frac{N\lambda}{2L} = 1 + \frac{(160)\left(600 \times 10^{-9}\ \text{m}\right)}{2\left(5.00 \times 10^{-2}\ \text{m}\right)} = \boxed{1.001}$$

P37.44 Counting light going both directions, the number of wavelengths originally in the cylinder is $m_1 = \frac{2L}{\lambda}$. It changes to $m_2 = \frac{2L}{\lambda/n} = \frac{2nL}{\lambda}$ as the cylinder is filled with gas. If N is the number of bright fringes passing, $N = m_2 - m_1 = \frac{2L}{\lambda}(n-1)$, or the index of refraction of the gas is

$$n = \boxed{1 + \frac{N\lambda}{2L}}$$

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Additional Problems

P37.45 The wavelength is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{60.0 \times 10^6 \text{ s}^{-1}} = 5.00 \text{ m}$$

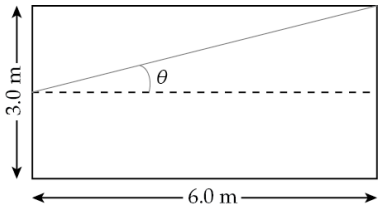
Along the line *AB* the two traveling waves going in opposite directions add to give a standing wave. The two transmitters are exactly 2.00 wavelengths apart and the signal from *B*, when it arrives at *A*, will always be in phase with transmitter *B.* Since *B* is 180° out of phase with *A*, the two signals always interfere destructively at the position of *A* to form a node.

The first antinode (point of constructive interference) is located at distance

$$\frac{\lambda}{4} = \frac{5.00 \text{ m}}{4} = \boxed{1.25 \text{ m}} \text{ from the node at } A$$

***P37.46** From ANS. FIG. P37.46, we note that the angle between the center line of the speakers and the corners of the room is

$$\theta = \tan^{-1}\left(\frac{1.5 \text{ m}}{6.0 \text{ m}}\right) = 14.0°$$



ANS. FIG. P37.46

In order for no other maxima to be heard, the $m = 1$ maximum must be more than 14.0° away from the central maximum. From Equation 37.2, the condition for constructive interference is

$$d \sin\theta_{\text{bright}} = m\lambda$$

or

$$\lambda = \frac{d \sin\theta_{\text{bright}}}{m} = \frac{v}{f}$$

where $v = 343$ m/s is the speed of sound. Solving for f and substituting $m = 1$ and $\theta = 14.0°$ then gives

$$f = \frac{v}{\lambda} = \frac{mv}{d \sin\theta_{\text{bright}}} = \frac{(1)(343 \text{ m/s})}{(1.0 \text{ m})\sin 14.0°} = \boxed{1.4 \times 10^2 \text{ Hz}}$$

P37.47 The same source will radiate light into the sugar solution with wavelength $\lambda_n = \frac{\lambda}{n}$. In other words, the condition for bright fringes becomes

$$d \sin\theta = m\lambda_n \rightarrow d \sin\theta = m\frac{\lambda}{n}$$

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Also, for small angles, as is the case here

$$\sin\theta \approx \tan\theta = \frac{y}{L}$$

The first side bright fringe ($m = 1$) is separated from the central bright fringe by distance y described by

$$d\sin\theta = m\frac{\lambda}{n} \quad \rightarrow \quad d\left(\frac{y}{L}\right) = \frac{\lambda}{n}$$

solving for y gives

$$y = \frac{\lambda L}{nd} = \frac{(560\times10^{-9}\ \text{m})(1.20\ \text{m})}{(1.38)(30.0\times10^{-6}\ \text{m})} = 1.62\times10^{-2}\ \text{m} = \boxed{1.62\ \text{cm}}$$

P37.48 (a) Where fringes of the two colors coincide we have

$$d\sin\theta = m\lambda = m'\lambda', \quad \text{requiring} \quad \frac{\lambda}{\lambda'} = \frac{m'}{m}$$

(b) $\lambda = 430\ \text{nm},\ \lambda' = 510\ \text{nm}$

$$\therefore \frac{m'}{m} = \frac{430\ \text{nm}}{510\ \text{nm}} = \frac{43}{51}$$

which cannot be reduced any further. Then $m = 51$, $m' = 43$. Then,

$$\theta_m = \sin^{-1}\left(\frac{m\lambda}{d}\right) = \sin^{-1}\left[\frac{(51)(430\times10^{-9}\ \text{m})}{0.025\times10^{-3}\ \text{m}}\right] = 61.3°$$

and

$$y_m = L\tan\theta_m = (1.5\ \text{m})\tan 61.3° = \boxed{2.74\ \text{m}}$$

P37.49 (a) Refer to ANS. FIG. P37.49. By similar triangles, the distance x between consecutive *like* interference fringes (bright-to-bright, or dark-to-dark) is to the change in thickness Δt of the air gap as the entire length of a plate ℓ (14.0 cm) is to the diameter d of the fiber (equal to the thickness of the air gap at the open end of the gap):

$$\frac{x}{\Delta t} = \frac{\ell}{d}$$

where, say, between consecutive destructive interference fringes

$$2t = \left(m + \frac{1}{2}\right)\lambda \quad \rightarrow \quad \Delta t = \frac{\lambda}{2}$$

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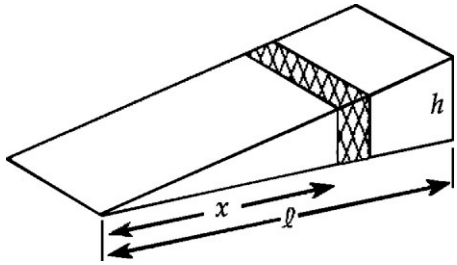
Combing the two relations gives

$$\frac{x}{\lambda/2} = \frac{\ell}{d}$$

and solving for the diameter d of the fiber then gives

$$d = \frac{\ell\lambda}{2x} = \frac{\left(14.0\times10^{-2}\text{ m}\right)\left(650\times10^{-9}\text{ m}\right)}{2\left(0.580\times10^{-3}\text{ m}\right)}$$

$$= 7.84\times10^{-5}\text{ m} = \boxed{78.4\ \mu\text{m}}$$



ANS. FIG. P37.49

P37.50 Assume the distance between gaps is 2 cm.

(a) Two adjacent directions of constructive interference for 600-nm light are described by $d\sin\theta = m\lambda$, with $\theta_0 = 0$. Then,

$$d\sin\theta = m\lambda$$

$$\left(2\times10^{-2}\text{ m}\right)\sin\theta_1 = 1\left(600\times10^{-9}\text{ m}\right)$$

Thus, $\theta_1 = 2\times10^{-3}\,^\circ$,

and $\theta_1 - \theta_0 \boxed{\sim 10^{-3}\,^\circ}$.

(b) We choose $\theta_1 = 20°$. Then,

$$\left(2\times10^{-2}\text{ m}\right)\sin 20° = (1)\lambda$$

Which gives $\lambda = 7$ mm. The frequency is then

$$f = \frac{c}{\lambda} = \frac{3\times10^{8}\text{ m/s}}{7\times10^{-3}\text{ m}} \boxed{\sim 10^{11}\text{ Hz}}$$

(c) Millimeter waves are $\boxed{\text{microwaves}}$.

P37.51 Constructive interference occurs where the phases of the waves differ by integral multiples m of 2π:

$$\left(\frac{2\pi x_1}{650} - 924\pi t + \frac{\pi}{6}\right) - \left(\frac{2\pi x_2}{650} - 924\pi t + \frac{\pi}{8}\right) = 2\pi m$$

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which becomes

$$\frac{2\pi(x_1 - x_2)}{650} + \left(\frac{\pi}{6} - \frac{\pi}{8}\right) = 2\pi m$$

$$\frac{(x_1 - x_2)}{650} + \frac{1}{12} - \frac{1}{16} = m$$

$$\boxed{x_1 - x_2 = \left(m - \tfrac{1}{48}\right)650, \text{ where } x_1 \text{ and } x_2 \text{ are in nanometers and } m = 0, 1, -1, 2, -2, 3, -3, \ldots}$$

P37.52 A bright line for the green light requires

$$d \sin\theta \approx d \tan\theta = m_1 \lambda_1$$

$$d\frac{y}{L} = m_1 \lambda_1$$

Similarly, a blue interference maximum requires

$$d\frac{y}{L} = m_2 \lambda_2$$

for integers m_1 and m_2. Thus,

$$m_1(540 \text{ nm}) = m_2(450 \text{ nm})$$

$$\frac{m_2}{m_1} = \frac{540 \text{ nm}}{450 \text{ nm}} = \frac{6}{5}$$

and smallest integers satisfying the equation are $m_1 = 5$ and $m_2 = 6$.

Then for both,

$$d\frac{y}{L} = 2\ 700 \text{ nm}$$

which gives

$$y = (2\ 700 \text{ nm})\frac{L}{d} = (2.7\ \mu\text{m})\left(\frac{1.4 \text{ m}}{150\ \mu\text{m}}\right) = \boxed{2.52 \text{ cm}}$$

P37.53 If the center point on the screen is to be a dark spot rather than bright, passage through the plastic must delay the light by one-half wavelength. Calling the thickness of the plastic t, $\frac{t}{\lambda} + \frac{1}{2} = \frac{t}{\lambda/n} = \frac{nt}{\lambda}$ or

$t = \boxed{\frac{\lambda}{2(n-1)}}$ where n is the index of refraction for the plastic.

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P37.54 There is no phase shift upon reflection from the upper surface (glass to air) of the film, but there will be a shift of $\frac{\lambda}{2}$ due to the reflection at the lower surface of the film (air to metal).

The total phase difference in the two reflected beams is then

$$\delta = 2nt + \frac{\lambda}{2}$$

For constructive interference, $\delta = m\lambda$, or

$$2(1.00)t + \frac{\lambda}{2} = m\lambda$$

Thus, the film thickness for the mth order bright fringe is

$$t_m = \left(m - \frac{1}{2}\right)\frac{\lambda}{2} = m\left(\frac{\lambda}{2}\right) - \frac{\lambda}{4}$$

and the thickness for the $m - 1$ bright fringe is:

$$t_{m-1} = (m-1)\left(\frac{\lambda}{2}\right) - \frac{\lambda}{4}$$

Therefore, the change in thickness required to go from one bright fringe to the next is

$$\Delta t = t_m - t_{m-1} = \frac{\lambda}{2}$$

To go through 200 bright fringes, the change in thickness of the air film must be

$$200\left(\frac{\lambda}{2}\right) = 100\lambda$$

Thus, the increase in the length of the rod is

$$\Delta L = 100\lambda = 100\left(5.00 \times 10^{-7}\ \text{m}\right) = 5.00 \times 10^{-5}\ \text{m}$$

From $\Delta L = L_i \alpha\, \Delta T$

we have: $\alpha = \dfrac{\Delta L}{L_i \Delta T} = \dfrac{5.00 \times 10^{-5}\ \text{m}}{(0.100\ \text{m})(25.0°\text{C})} = \boxed{20.0 \times 10^{-6}°\text{C}^{-1}}$

P37.55 Since $1 < 1.25 < 1.34$, light reflected from top and bottom surfaces of the oil undergoes phase reversal. The path difference is then $2t$, thus

$$2t = m\lambda_n = \frac{m\lambda}{n}$$

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for maximum reflection, with $m = 1$ for the given first-order condition and $n = 1.25$. So

$$t = \frac{m\lambda}{2n} = \frac{1(500\text{ nm})}{2(1.25)} = 200\text{ nm}$$

The volume we assume to be constant:

$$1.00\text{ m}^3 = (200\text{ nm})A$$

The area is then

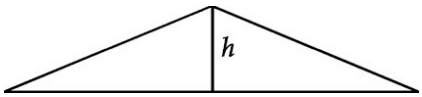
$$A = \frac{1.00\text{ m}^3}{(200\times10^{-9}\text{ m})} = 5.00\times10^{6}\text{ m}^2 = \boxed{5.00\text{ km}^2}$$

P37.56 The interfering waves travel either along the hypotenuses or the bases of the right triangles. The total length of the two bases is 15.0 km. The condition for destructive interference for minimum height h is

$$2\sqrt{(15.0\times10^{3}\text{ m})^2 + h^2} - 2(15.0\times10^{3}\text{ m}) = \lambda/2 = 175\text{ m}$$

$$2\sqrt{(15.0\times10^{3}\text{ m})^2 + h^2} = 30.175\times10^{3}\text{ m}$$

$$h = 1.62\times10^{3}\text{ m} = \boxed{1.62\text{ km}}$$



ANS. FIG. P37.56

P37.57 We may treat this problem as a double slit experiment where the second slit is the mirror image of the source, 1.00 cm below the mirror plane; however, we must remember that the light undergoes a $\frac{\pi}{2}$ phase shift at the mirror, so light and dark fringes are interchanged in the interference pattern. Thus, for destructive interference, the path length must differ by $m\lambda$. For dark for the first dark fringe (modifying Equation 37.7), we have

$$y_{\text{dark}} = \frac{m\lambda L}{d} = \frac{1(5.00\times10^{-7}\text{ m})(100\text{ m})}{(2.00\times10^{-2}\text{ m})} = \boxed{2.50\text{ mm}}$$

P37.58 From Equation 37.14, for wavelength $\lambda_1 = 600\text{ nm}$,

$$\frac{I}{I_{\max}} = \cos^2\left(\frac{\pi yd}{\lambda_1 L}\right) = 0.810$$

$$\frac{\pi yd}{L} = \lambda_1\cos^{-1}\left(\frac{I_1}{I_{\max}}\right)^{1/2} = (600\text{ nm})\cos^{-1}(0.810)^{1/2} = 271\text{ nm}$$

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For the same y, d, and L, let λ_2 be the wavelength for which

$$\frac{I_2}{I_{2,\max}} = 0.640$$

Then,

$$\lambda_2 = \frac{\pi yd/L}{\cos^{-1}\left(I_2/I_{2,\max}\right)^{1/2}} = \frac{271 \text{ nm}}{\cos^{-1}(0.640)^{1/2}} = \boxed{421 \text{ nm}}$$

Note that in this problem, $\cos^{-1}\left(\frac{I}{I_{\max}}\right)^{1/2}$ must be expressed in radians.

P37.59 As with any air gap between glass plates, light reflecting off the lower plate undergoes a phase reversal. Thus, for the mth-dark fringe after the first fringe ($m = 0$), with the gap filled with air:

$$2nt = m\lambda$$

where $n = 1.00$ and $m = 1, 2, \ldots, 84$. So, at the widest edge of the wedge,

$$t = \frac{84\lambda}{2} = 42\lambda$$

When submerged in water,

$$2nt = m\lambda$$

$$m = \frac{2nt}{\lambda} = \frac{2(1.33)\left[(42)\lambda\right]}{\lambda} = 111.7 = 111$$

So, counting the first fringe ($m = 0$), the total number of fringes is

$m + 1 =$ $\boxed{112 \text{ dark fringes}}$

P37.60 Refer to Figure P37.60. Call t the thickness of the sheet. With the sheet in place, the central maximum corresponds to zero phase difference. Thus, the added distance δ traveled by the light from the lower slit introduces a phase difference equal to that introduced by the plastic film sheet. Call the original length of the path from the upper slit to the screen D; then, the original number of wavelengths along distance D are

$$N_0 = \frac{D}{\lambda_a}$$

where λ_a is the wavelength in air. With the plastic sheet in the path, the number of wavelengths changes to

$$N = \frac{D-t}{\lambda_a} + \frac{t}{\lambda_p} = \frac{D-t}{\lambda_a} + \frac{t}{\lambda_a/n} = \frac{D-t+nt}{\lambda_a} = \frac{D+(n-1)t}{\lambda_a}$$

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where λ_a is the wavelength in plastic. The phase difference introduced by the plastic sheet is

$$\delta\phi = 2\pi\left(N - N_0\right) = 2\pi\left[\frac{D+(n-1)t}{\lambda_a} - \frac{D}{\lambda_a}\right] = 2\pi\frac{(n-1)t}{\lambda_a}$$

The corresponding difference in **path length** δ is

$$\delta = \delta\phi\left(\frac{\lambda_a}{2\pi}\right) = \left[2\pi\frac{(n-1)t}{\lambda_a}\right]\left(\frac{\lambda_a}{2\pi}\right) = t(n-1)$$

Note that the wavelength of the light does not appear in this equation. In the figure, the two rays from the slits are essentially parallel.

Thus the angle θ may be expressed as

$$\sin\theta = \frac{\delta}{d} = \frac{(n-1)t}{d} \quad \rightarrow \quad \theta = \sin^{-1}\left[\frac{(n-1)t}{d}\right]$$

The height y of the central maximum is given by

$$\frac{y'}{L} = \tan\theta$$

from which we obtain

$$y = \boxed{L\tan\left\{\sin^{-1}\left[\frac{(n-1)t}{d}\right]\right\} = \frac{(n-1)Lt}{\sqrt{d^2 - (n-1)^2 t^2}}}$$

P37.61 From Figure P37.61, observe that the distance that the ray travels from the top of the transmitter to the ground is

$$x = \sqrt{h^2 + \left(\frac{d}{2}\right)^2}$$

$$= \sqrt{(35.0\text{ m})^2 + \left(\frac{50.0\text{ m}}{2}\right)^2} = \sqrt{1850\text{ m}^2} = 43.0\text{ m}$$

Including the phase reversal due to reflection from the ground, the total shift between the two waves (transmitter-to-ground-to-receiver and transmitter-to-receiver) is

$$\delta = 2x + \frac{\lambda}{2} - d$$

For constructive interference,

$$2x + \frac{\lambda}{2} - d = m\lambda \;\rightarrow\; \lambda = \frac{2x-d}{\left(m - \frac{1}{2}\right)}$$

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and for destructive interference

$$2x+\frac{\lambda}{2}-d=\left(m+\frac{1}{2}\right)\lambda \;\rightarrow\; \lambda=\frac{2x-d}{m}$$

(a) The longest wavelength that interferes constructively is, for $m = 1$,

$$\lambda=\frac{2x-d}{\left(1-\frac{1}{2}\right)}=14x-2d=4\sqrt{1850\ \text{m}^2}-2(50.0\ \text{m})=\boxed{72.0\ \text{m}}$$

(b) The longest wavelength that interferes destructively is, for $m = 1$,

$$\lambda=\frac{2x-d}{1}=2\sqrt{1850\ \text{m}^2}-50.0\ \text{m}=\boxed{36.0\ \text{m}}$$

P37.62 From Figure P37.57, observe that the distance that the ray travels from the top of the transmitter to the ground is

$$x=\sqrt{h^2+\left(\frac{d}{2}\right)^2}=\frac{\sqrt{4h^2+d^2}}{2}$$

Including the phase reversal due to reflection from the ground, the total shift between the two waves (transmitter-to-ground-to-receiver and transmitter-to-receiver) is

$$\delta=2x+\frac{\lambda}{2}-d$$

For constructive interference,

$$2x+\frac{\lambda}{2}-d=m\lambda \;\rightarrow\; \lambda=\frac{2x-d}{\left(m-\frac{1}{2}\right)}$$

and for destructive interference

$$2x+\frac{\lambda}{2}-d=\left(m+\frac{1}{2}\right)\lambda \;\rightarrow\; \lambda=\frac{2x-d}{m}$$

(a) The longest wavelength that interferes constructively is, for $m = 1$,

$$\lambda=\frac{2x-d}{\left(1-\frac{1}{2}\right)}=4x-2d=\frac{4\sqrt{4h^2+d^2}}{2}-2d=\boxed{2\sqrt{4h^2+d^2}-2d}$$

(b) The longest wavelength that interferes destructively is, for $m = 1$,

$$\lambda=\frac{2x-d}{1}=\boxed{\sqrt{4h^2+d^2}-d}$$

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P37.63 (a) There is a phase reversal by reflection at the flat plate. Constructive interference in the reflected light requires $2t = \left(m + \frac{1}{2}\right)\lambda$.

The first bright ring has $m = 0$ and the 55th has $m = 54$, so at the edge of the lens

$$t = \left(m + \frac{1}{2}\right)\frac{\lambda}{2} = (54.5)\frac{650 \times 10^{-9}\ \text{m}}{2} = 17.7\ \mu\text{m}$$

Now from the geometry in textbook Figure P37.59, we can find the distance t from the curved surface down to the flat plate by considering distances measured from the center of curvature:

$$\sqrt{R^2 - r^2} = R - t \quad \text{or} \quad R^2 - r^2 = R^2 - 2Rt + t^2$$

Solving for R gives

$$R = \frac{r^2 + t^2}{2t} = \frac{\left(5.00 \times 10^{-2}\ \text{m}\right)^2 + \left(1.77 \times 10^{-5}\ \text{m}\right)^2}{2\left(1.77 \times 10^{-5}\ \text{m}\right)} = \boxed{70.6\ \text{m}}$$

(b) $$\frac{1}{f} = (n-1)\left(\frac{1}{R_2} - \frac{1}{R_2}\right) = 0.520\left(\frac{1}{\infty} - \frac{1}{-70.6\ \text{m}}\right) \quad \text{so} \quad f = \boxed{136\ \text{m}}$$

P37.64 Reflection off the top surface of the wedge produced a phase reversal, but light reflecting off the bottom surface produces no phase change. Thus, a first *dark* fringe occurs at the thin end of the wedge. For bright fringes in the thin film, the thickness is given by Equation 37.17:

$$t = \frac{\left(m + \frac{1}{2}\right)\lambda}{2n}$$

The first fringe corresponds to $m = 0$, the second to $m = 1$, etc.; so the Nth fringe corresponds to $N = m + 1$.

To find how many fringes are present, we solve for m by setting $t = h$:

$$m + \frac{1}{2} = \frac{2nt}{\lambda} = \frac{2nh}{\lambda} = \frac{2(1.50)\left(1.00 \times 10^{-3}\ \text{m}\right)}{\left(632.8 \times 10^{-9}\ \text{m}\right)} = 4\ 740$$

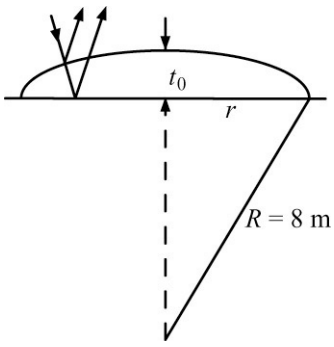
$$\therefore \quad m = 4\ 740$$

So, the number of fringes is $N = m + 1 = 4\ 741$. This number is less than 5000.

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P37.65 Light reflecting from the upper interface of the air layer suffers no phase change, while light reflecting from the lower interface is reversed 180°. Then there is indeed a dark fringe at the outer circumference of the lens, and a dark fringe wherever the air thickness t satisfies

$$2t = m\lambda, \qquad m = 0, 1, 2, \ldots.$$



ANS. FIG. P37.65

(a) At the central dark spot, $m = 50$ and

$$t_0 = \frac{50\lambda}{2}$$

$$= 25\left(589\times10^{-9}\text{ m}\right) = 1.47\times10^{-5}\text{ m} = \boxed{14.7\ \mu\text{m}}$$

(b) In the right triangle,

$$R^2 = r^2 + \left(R - t_0\right)^2$$

$$(8.00\text{ m})^2 = r^2 + \left(8.00\text{ m} - 1.47\times10^{-5}\text{ m}\right)^2$$

$$\cancel{(8.00\text{ m})^2} = r^2 + \cancel{(8.00\text{ m})^2} - 2(8.00\text{ m})\left(1.47\times10^{-5}\text{ m}\right) + 2.16\times10^{-10}\text{ m}^2$$

$$r^2 = 2(8.00\text{ m})\left(1.47\times10^{-5}\text{ m}\right) - 2.16\times10^{-10}\text{ m}^2$$

The last term is negligible. Then,

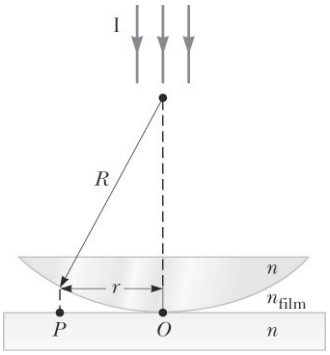
$$r = \sqrt{2(8\text{ m})\left(1.47\times10^{-5}\text{ m}\right)} = 1.53\times10^{-2}\text{ m} = \boxed{1.53\text{ cm}}$$

(c) $$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.50-1)\left(\frac{1}{\infty} - \frac{1}{8.00\text{ m}}\right)$$

$$\boxed{f = -16.0\text{ m}}$$

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P37.66 The shift between the waves reflecting from the top and bottom surfaces of the film at the point where the film has thickness t is $\delta = 2tn_{\text{film}} + \frac{\lambda}{2}$, with the factor of $\frac{\lambda}{2}$ being due to a phase reversal at *one* of the surfaces.



ANS. FIG. P37.66

For the dark rings (destructive interference), the total shift should be $\delta = \left(m + \frac{1}{2}\right)\lambda$ with $m = 0, 1, 2, 3, \ldots$. This requires that $t = \frac{m\lambda}{2n_{\text{film}}}$. To find t in terms of r and R,

$$R^2 = r^2 + (R - t)^2 \rightarrow r^2 = 2Rt + t^2$$

Since t is much smaller than R, $t^2 << 2Rt$, therefore

$$r^2 \approx 2Rt = 2R\left(\frac{m\lambda}{2n_{\text{film}}}\right)$$

Thus, $\boxed{r \approx \sqrt{\frac{m\lambda R}{n_{\text{film}}}}}$ where m is an integer.

P37.67 Refer to the solution of P37.57. We may treat this as a double-slit interference problem, where $d = 2h$, but with maxima and minima interchanged because of phase reversal caused by the reflection off the mirror:

$$d\sin\theta = 2h\sin\theta = \left(m + \frac{1}{2}\right)\lambda \quad \text{bright fringe}$$

and $\sin\theta \approx \tan\theta = \frac{y}{L}$ for small angles; hence,

$$2h\sin\theta = \left(m + \frac{1}{2}\right)\lambda$$

$$2h\left(\frac{y}{L}\right) = \left(m + \frac{1}{2}\right)\lambda$$

The spacing between consecutive fringes corresponding to m and $m + 1$ is

$$2h\left(\frac{\Delta y}{L}\right) = \lambda$$

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so

$$h = \frac{L\lambda}{2\Delta y} = \frac{(2.00\text{ m})\left(606\times10^{-9}\text{ m}\right)}{2\left(1.20\times10^{-3}\text{ m}\right)}$$

$$= 5.05\times10^{-4}\text{ m} = \boxed{0.505\text{ mm}}$$

P37.68 (a) For a linear function taking the value $n = 1.90$ at $y = 0$ and $n = 1.33$ at $y = 20.0$ cm, we write

$$n(y) = 1.90 + (1.33 - 1.90)y/(20.0\text{ cm})$$

or $\boxed{n(y) = 1.90 - 0.0285\,y/\text{cm}}$

(b) The optical path length is

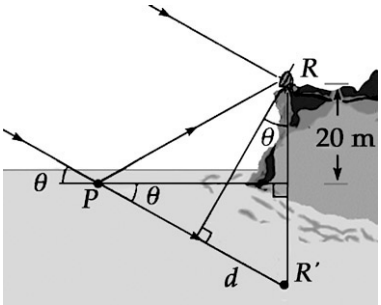
$$\int_0^{20.0\text{ cm}} n(y)\,dy = \int_0^{20.0\text{ cm}} [1.90 - 0.0285\,y/\text{cm}]\,dy$$

$$= 1.90y - \frac{0.0285y^2}{2}\Bigg|_0^{20.0\text{ cm}}$$

$$= 38.0\text{ cm} - 5.7\text{ cm} = \boxed{32.3\text{ cm}}$$

(c) A wavefront slows down as it travels deeper into the mixture to regions of greater index of refraction. The lower part of the wavefront travels more slowly than the upper part; the result is that the wavefront bends, becoming more horizontal. The path is similar to that of a beam crossing the boundary between a medium of lesser to a medium of greater index of refraction, as, for example, from air into water: the beam tends to bend toward the normal. The difference is that the change in direction is gradual rather than sudden. The beam will continuously curve downward.

P37.69 One radio wave reaches the receiver R directly from the distant source at an angle θ above the horizontal. The other wave undergoes phase reversal as it reflects from the water at P. The distance from P to R is the same as from P to R', where R' is the mirror image of the telescope. Therefore, the path difference is d.



ANS. FIG. P37.69

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Constructive interference first occurs for a path difference of

$$d = \frac{\lambda}{2} \qquad \textbf{[1]}$$

The angles θ in the figure are equal because they each form part of a right triangle with a shared angle at *R'*.

So the path difference is

$$d = 2(20.0\text{ m})\sin\theta = (40.0\text{ m})\sin\theta$$

The wavelength is

$$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\text{ m/s}}{60.0\times10^{6}\text{ Hz}} = 5.00\text{ m}$$

Substituting for d and λ in equation [1],

$$(40.0\text{ m})\sin\theta = \frac{5.00\text{ m}}{2}$$

Solving for the angle θ,

$$\theta = \sin^{-1}\left(\frac{5.00\text{ m}}{80.0\text{ m}}\right) = \boxed{3.58°}$$

P37.70 One phase reversal occurs by reflection off the front of the soap film.

(a) Bright bands are observed when $2nt = \left(m + \frac{1}{2}\right)\lambda$.

Hence, the first bright band ($m = 0$) corresponds to $nt = \frac{\lambda}{4}$.

By similar triangles, the distance x from the top where a fringe occurs is proportional to the thickness t of the film:

$$\frac{x_1}{x_2} = \frac{t_1}{t_2}$$

Thus, we have

$$x_2 = x_1\left(\frac{t_2}{t_1}\right) = x_1\left(\frac{\lambda_2}{\lambda_1}\right) = (3.00\text{ cm})\left(\frac{680\text{ nm}}{420\text{ nm}}\right) = \boxed{4.86\text{ cm}}$$

(b) $t_1 = \frac{\lambda_1}{4n} = \frac{420\text{ nm}}{4(1.33)} = \boxed{78.9\text{ nm}}$

$t_2 = \frac{\lambda_2}{4n} = \frac{680\text{ nm}}{4(1.33)} = \boxed{128\text{ nm}}$

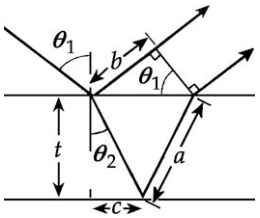
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(c) $$\theta \approx \tan\theta = \frac{t_1}{x_1} = \frac{78.9 \text{ nm}}{3.00 \text{ cm}} = \boxed{2.63 \times 10^{-6} \text{ rad}}$$

Challenge Problems

P37.71 Refer to ANS. FIG. P37.71 for the geometry of the situation. At the air-film interface, Snell's law gives

$$1.00 \sin 30.0° = 1.38 \sin\theta_2 \quad \rightarrow \quad \theta_2 = 21.2°$$



ANS. FIG. P37.71

Call t the unknown thickness of the film. Then,

$$\cos 21.2° = \frac{t}{a} \quad \rightarrow \quad a = \frac{t}{\cos 21.2°}$$

$$\tan 21.2° = \frac{c}{t} \quad \rightarrow \quad c = t \tan 21.2°$$

$$\sin\theta_1 = \frac{b}{2c} \quad \rightarrow \quad b = 2t(\tan 21.2°)(\sin 30.0°)$$

The net shift for the second ray, including the phase reversal on reflection of the first, is

$$2an - b - \frac{\lambda}{2}$$

where the factor n accounts for the shorter wavelength in the film. For constructive interference, we require

$$2an - b - \frac{\lambda}{2} = m\lambda$$

The minimum thickness will occur when $m = 0$ and will be given by

$$2an - b - \frac{\lambda}{2} = 0$$

Then,

$$\frac{\lambda}{2} = 2an - b = 2\frac{nt}{\cos 21.2°} - 2t(\tan 21.2°)(\sin 30.0°)$$

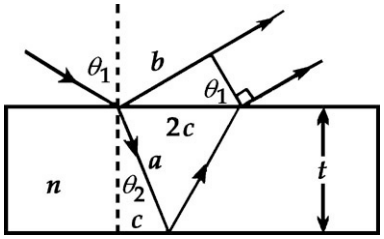
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and $$\frac{590\text{ nm}}{2} = \left[\frac{2(1.38)}{\cos 21.2°} - 2(\tan 21.2°)(\sin 30.0°)\right] t = 2.57t$$

which gives $t = \boxed{115\text{ nm}}$.

P37.72 The shift between the two reflected waves is $\delta = 2na - b - \frac{\lambda}{2}$, where a and b are as shown in the ray diagram in ANS. FIG. P37.72, n is the index of refraction, and the term $\frac{\lambda}{2}$ is due to phase reversal at the top surface. For constructive interference, $\delta = m\lambda$, where m has integer values. This condition becomes

$$2na - b = \left(m + \frac{1}{2}\right)\lambda \qquad \textbf{[1]}$$



ANS. FIG. P37.72

From the figure's geometry,

$$a = \frac{t}{\cos\theta_2}$$

$$c = a\sin\theta_2 = \frac{t\sin\theta_2}{\cos\theta_2}$$

$$b = 2c\sin\theta_1 = \frac{2t\sin\theta_2}{\cos\theta_2}\sin\theta_1$$

Also, from Snell's law, $\sin\theta_1 = n\sin\theta_2$.

Thus, $$b = \frac{2nt\sin^2\theta_2}{\cos\theta_2}.$$

With these results, the condition for constructive interference given in equation [1] becomes:

$$2n\left(\frac{t}{\cos\theta_2}\right) - \frac{2nt\sin^2\theta_2}{\cos\theta_2} = \left(m + \frac{1}{2}\right)\lambda$$

$$\frac{2nt}{\cos\theta_2}\left(1 - \sin^2\theta_2\right) = \left(m + \frac{1}{2}\right)\lambda$$

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$$2nt\frac{\left(1-\sin^2\theta_2\right)}{\sqrt{1-\sin^2\theta_2}}=\left(m+\frac{1}{2}\right)\lambda$$

or $$2nt\sqrt{1-\sin^2\theta_2}=\left(m+\frac{1}{2}\right)\lambda$$

Using $\sin\theta_1 = n\sin\theta_2 \rightarrow \sin\theta_2 = \sin\theta_1/n$, we have finally

$$\boxed{2nt\sqrt{1-\frac{\sin^2\theta_1}{n^2}}=\left(m+\tfrac{1}{2}\right)\lambda, \text{ where } m=0, 1, 2, \ldots}$$

P37.73 (a) Minimum: $2nt = m\lambda_2$ for $m = 0, 1, 2, \ldots$

Maximum: $2nt=\left(m'+\frac{1}{2}\right)\lambda_1$ for $m' = 0, 1, 2, \ldots$

Note that m and m' are distinct integer values, and must be consecutive because no intensity minima are observed between λ_1 and λ_2.

Also, $\lambda_1 > \lambda_2 \rightarrow \left(m'+\frac{1}{2}\right) < m$, so $m' = m - 1$.

Thus, we have

$$2nt = m\lambda_2 = \left(m'+\frac{1}{2}\right)\lambda_1 = \left[(m-1)+\frac{1}{2}\right]\lambda_1$$

$$m\lambda_2 = \left(m-\frac{1}{2}\right)\lambda_1$$

$$2m\lambda_2 = 2m\lambda_1 - \lambda_1$$

so $$\boxed{m=\frac{\lambda_1}{2\left(\lambda_1-\lambda_2\right)}}.$$

(b) $$m=\frac{500\,\text{nm}}{2\left(500\,\text{nm}-370\,\text{nm}\right)}=1.92 \rightarrow 2$$ (wavelengths measured to ± 5 nm)

Minimum: $2nt = m\lambda_2$

$2(1.40)t = 2(370\text{ nm})$ $t = 264$ nm

Maximum: $2nt=\left(m'+\frac{1}{2}\right)\lambda=\left(m-1+\frac{1}{2}\right)\lambda=1.5\lambda$

$2(1.40)t = 1.5(500\text{ nm}) \rightarrow$ $t = 268$ nm

Film thickness = $\boxed{266\text{ nm}}$

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P37.74 The amplitude of the light from slit 1 is three times that from slit 2; therefore, the magnitude of the light arriving at the screen at some point P is

$$\begin{aligned} E_P &= E_1 + E_2 = 3E_0 \sin(\omega t) + E_0 \sin(\omega t + \phi) \\ &= E_0\left[3\sin\omega t + \sin(\omega t + \phi)\right] \\ \frac{E_P}{E_0} &= 3\sin(\omega t) + \sin(\omega t + \phi) \\ &= 3\sin(\omega t) + \left[\sin(\omega t)\cos(\phi) + \cos(\omega t)\sin(\phi)\right] \\ &= \sin(\omega t)\left[3 + \cos(\phi)\right] + \cos(\omega t)\sin(\phi) \end{aligned}$$

The square of this expression is

$$\begin{aligned} \left(\frac{E_P}{E_0}\right)^2 &= \sin^2(\omega t)\left[3 + \cos(\phi)\right]^2 \\ &\quad + 2\sin(\omega t)\cos(\omega t)\left[3 + \cos(\phi)\right]\sin(\phi) \\ &\quad + \cos^2(\omega t)\sin^2(\phi) \\ \left(\frac{E_P}{E_0}\right)^2 &= \sin^2(\omega t)\left[3 + \cos(\phi)\right]^2 + \sin(2\omega t)\left[3 + \cos(\phi)\right]\sin(\phi) \\ &\quad + \cos^2(\omega t)\sin^2(\phi) \end{aligned}$$

and the time average of this expression is

$$\begin{aligned} \overline{\left(\frac{E_P}{E_0}\right)^2} &= \frac{1}{2}\left[3 + \cos(\phi)\right]^2 + \frac{1}{2}\sin^2(\phi) \\ &= \frac{1}{2}\left[9 + 6\cos(\phi) + \cos^2(\phi) + \sin^2(\phi)\right] = \frac{1}{2}\left[10 + 6\cos(\phi)\right] \end{aligned}$$

because the time average of $\sin^2(\omega t)$ and $\cos^2(\omega t)$ is $\frac{1}{2}$, and the time average of $\sin(2\omega t)$ is zero. Using the identity

$$\cos(\phi) = \cos\left(\frac{\phi}{2} + \frac{\phi}{2}\right) = 2\cos^2\left(\frac{\phi}{2}\right) - 1$$

we have

$$\begin{aligned} \overline{\left(\frac{E_P}{E_0}\right)^2} &= \frac{1}{2}\left[10 + 6\cos(\phi)\right] = \frac{1}{2}\left[10 + 6\left(2\cos^2\left(\frac{\phi}{2}\right) - 1\right)\right] \\ &= \frac{1}{2}\left[4 + 12\cos^2\left(\frac{\phi}{2}\right)\right] = 2\left[1 + 3\cos^2\left(\frac{\phi}{2}\right)\right] \end{aligned}$$

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Intensity is proportional to the time average of the square of the amplitude, so

$$I \propto \overline{E_P^2} = 2E_0^2\left[1+3\cos^2\left(\frac{\phi}{2}\right)\right]$$

At the central maximum, $\phi = 0$, so the maximum intensity is

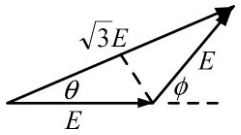
$$I_{\max} \propto 2E_0^2\left[1+3\cos^2(0)\right] = 2E_0^2(4) = 8E_0^2$$

Thus, we have

$$\frac{I}{I_{\max}} = \frac{2E_0^2\left[1+3\cos^2\left(\frac{\phi}{2}\right)\right]}{8E_0^2} = \frac{1}{4}\left[1+3\cos^2\left(\frac{\phi}{2}\right)\right]$$

$$I = \frac{I_{\max}}{4}\left[1+3\cos^2\left(\frac{\phi}{2}\right)\right]$$

P37.75 Represent the light radiated from each slit to point P as a phasor. The two have very nearly equal amplitudes E. Since intensity is proportional to amplitude squared, we are told they add to amplitude $\sqrt{3}E$. As shown in the figure, the triangle representing the sum of phasors may be divided into two right triangles whose common side that bisects the line of length $\sqrt{3}E$. From either triangle, we see that



ANS. FIG. P37.75

$$\cos\theta = \frac{\sqrt{3}\,E/2}{E} \quad \rightarrow \quad \theta = 30°$$

Next, the obtuse angle between the two phasors is 180 – 30 – 30 = 120°, and so $\phi = 180 - 120° = 60°$.

The phase difference between the two phasors is caused by the path difference from S to the slits, $\delta = \overline{SS_2} - \overline{SS_1}$, according to $\frac{\delta}{\lambda} = \frac{\phi}{360°}$, $\delta = \lambda\frac{60°}{360°} = \frac{\lambda}{6}$. Then

$$\delta = \sqrt{L^2+d^2} - L = \frac{\lambda}{6}$$

$$L^2 + d^2 = L^2 + \frac{2L\lambda}{6} + \frac{\lambda^2}{36} \rightarrow d^2 = \frac{2L\lambda}{6} + \frac{\lambda^2}{36}$$

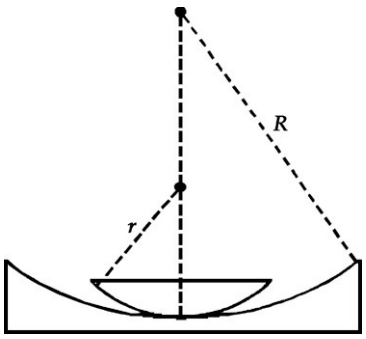
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The last term is negligible, so

$$d = \left(\frac{2L\lambda}{6}\right)^{1/2} = \sqrt{\frac{2(1.2\text{ m})\left(620\times10^{-9}\text{ m}\right)}{6}} = \boxed{0.498\text{ mm}}$$

P37.76 For bright rings the gap t between surfaces is given by $2t = \left(m + \frac{1}{2}\right)\lambda$.

The first bright ring has $m = 0$ and the hundredth has $m = 99$.



ANS. FIG. P37.76

So, $$t = \frac{1}{2}(99.5)\left(500\times10^{-9}\text{ m}\right) = 24.9\ \mu\text{m}$$

Call r_b the ring radius. From the geometry shown in ANS. FIG. P37.76,

$$t = \left(r - \sqrt{r^2 - r_b^2}\right) - \left(R - \sqrt{R^2 - r_b^2}\right)$$

$$= r - r\sqrt{1 - \left(\frac{r_b}{r}\right)^2} - R + R\sqrt{1 - \left(\frac{r_b}{R}\right)^2}$$

Since $r_b << r$, we can expand in binomial series:

$$t = r - r\left(1 - \frac{1}{2}\frac{r_b^2}{r^2}\right) - R + R\left(1 - \frac{1}{2}\frac{r_b^2}{R^2}\right) = \frac{1}{2}\frac{r_b^2}{r} - \frac{1}{2}\frac{r_b^2}{R}$$

$$r_b = \left[\frac{2t}{1/r - 1/R}\right]^{1/2} = \left[\frac{2\left(24.9\times10^{-6}\text{ m}\right)}{1/4.00\text{ m} - 1/12.0\text{ m}}\right]^{1/2} = \boxed{1.73\text{ cm}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P37.2 3.53 mm

P37.4 515 nm

P37.6 The sine of the angle for $m = 1$ fringe is greater than 1, which is impossible.

P37.8 (a) 1.77 μm; (b) 1.47 μm

P37.10 36.2 cm

P37.12 (a) 34.9°; (b) 5.25 cm; (c) 5.24×10^{14} Hz

P37.14 11.3 m

P37.16 $v \tan\left[\sin^{-1}\left(\frac{m\lambda}{d}\right)\right]$

P37.18 (a) 13.2 rad; (b) 6.28 rad; (c) 1.27×10^{-2} deg; (d) 5.97×10^{-2} deg

P37.20 (a) 22.6 cm; (b) 2.51×10^{-3}; (c) 6.03×10^{-7} m; (d) 7.21°; (e) 2.28 cm; (f) The two answers are close but do not agree exactly. The fringes are not laid out linearly on the screen as assumed in part (a), and this nonlinearity is evident for relatively large angles such as 7.21°.

P37.22 (a) 10 m; (b) 500 m; (c) See P37.22(c) for full explanation.

P37.24 E_R = 10.0 and $\phi = 53.1°$

P37.26 (a) $I = \frac{I_{\max}}{9}\left[1 + 2\cos\left(\frac{2\pi d \sin\theta}{\lambda}\right)\right]^2$; (b) See P37.24(b) for full explanation; (c) 9:1

P37.28 See ANS. FIG. P37.28.

P37.30 (a) 638 nm; (b) A thicker film would require a higher order of reflection, so use a larger value of m; (c) 360 nm, 600 nm

P37.32 96.2 nm

P37.34 (a) 276 nm, 138 nm, 92.0 nm; (b) No visible wavelengths are intensified.

P37.36 (a) green; (b) violet

P37.38 (a) 238 nm; (b) λ increase; (c) 328 nm

P37.40 (a) 97.8 nm; (b) Yes. Destructive interference occurs when $2nt = (m + \frac{1}{2})\lambda$ (Eq. 37.17), where m is an integer. (There is a phase change at both faces of the film in Figure P37.40.) Hence, for m = 1, 2,… we obtain thicknesses of 293 nm, 489 nm, . . .

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P37.42 39.6 μm

P37.44 $1+\frac{N\lambda}{2L}$

P37.46 1.4×10^{2} Hz

P37.48 (a) See P37.48(a) for full explanation; (b) 2.74 m

P37.50 (a) ~10^{-3} degree; (b) ~10^{11} Hz; (c) microwaves

P37.52 2.52 cm

P37.54 $20.0 \times 10^{-6}\ ^{\circ}\text{C}^{-1}$

P37.56 1.62 km

P37.58 421 nm

P37.60 $y = L\tan\left\{\sin^{-1}\left[\frac{(n-1)t}{d}\right]\right\} = \frac{(n-1)Lt}{\sqrt{d^2-(n-1)^2t^2}}$

P37.62 (a) $2\sqrt{4h^2+d^2}-2d$; (b) $\sqrt{4h^2+d^2}-d$

P37.64 The number of fringes is $N = m + 1 = 474\ 1$. This number is less than 5 000.

P37.66 $r \approx \sqrt{\frac{m\lambda R}{n_{\text{film}}}}$

P37.68 (a) $n(y) = 1.90 - 0.0285\ y/\text{cm}$; (b) 32.3 cm; (c) The beam will continuously curve downward.

P37.70 (a) 4.86 cm; (b) 78.9 nm, 128 nm; (c) 2.63×10^{-6} rad

P37.72 $2nt\sqrt{1-\frac{\sin^2\theta_1}{n^2}} = \left(m+\frac{1}{2}\right)\lambda,$ where $m = 0, 1, 2, \ldots$

P37.74 See P37.74 for full explanation.

P37.76 1.73 cm

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